An Enterprise-Grade Explainable Multimodal AI Platform for Remote Epilepsy Care

Design, Implementation, and Evaluation

A dissertation submitted in partial fulfilment of the requirements for the degree of

Doctor of Business Administration (DBA)

Scope: Epilepsy · Reference index case: Patient EP001

Empirical basis: real scalp EEG (CHB-MIT) + external validation (EEG-Eye-State)

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> ***Compiled PDF:*** *this dissertation is compiled to DBA-Epilepsy-Thesis.pdf (52 pp) via* `python analysis/build_thesis_pdf.py`*. The companion technical deliverable is DBA-Epilepsy-Deliverable.pdf.*

Abstract

Epilepsy affects approximately fifty million people worldwide, yet its diagnosis and ongoing management remain slow, fragmented, and unevenly explained. Clinical history captured by neurologists and electrophysiological evidence captured by electroencephalography (EEG) technicians are rarely fused within a single, governed, enterprise workflow, and the machine-learning models that could assist are too often delivered as isolated artefacts without the data-engineering, deployment, monitoring, and governance scaffolding that a healthcare organisation requires. This dissertation addresses that organisational gap. Adopting a Design Science Research paradigm combined with a mixed-methods evaluation, it designs, implements, and evaluates an enterprise-grade, explainable, multimodal artificial-intelligence platform for remote epilepsy care, structured as seven connected operating pipelines spanning research protocol, data engineering, feature engineering, statistical and machine learning modelling, machine-learning operations, retrieval-augmented generation with a knowledge graph, and clinical safety with responsible-AI governance.

The empirical core of the work is a reproducible analysis of **real scalp EEG** from the CHB-MIT database. Using signal preprocessing, time-frequency transforms (short-time Fourier and continuous wavelet), one-dimensional-to-two-dimensional image representations, and a set of interpretable biomarkers (spectral band power, Hjorth parameters, Higuchi fractal dimension, spectral entropy, phase-locking value, and line-length), a hyperparameter-tuned, class-balanced classifier discriminates ictal from interictal states with a cross-validated area under the receiver-operating-characteristic curve (ROC-AUC) of approximately 0.92, rising to approximately 0.97 for ictal-versus-interictal separation, and generalises to an external EEG benchmark (EEG-Eye-State) at a ROC-AUC of approximately
0.979. Explainability via SHAP and permutation importance converges on line-length, gamma-band power, and phase-locking value as the dominant discriminators, and non-parametric hypothesis testing confirms their significance. Clinical severity, seizure-recurrence, and longitudinal models are demonstrated on

a clearly labelled synthetic 500-patient cohort pending institutional-review-board-approved clinical data. The platform is delivered as running code, an interactive viewer with fifteen analytical views, a knowledge graph exported as Resource Description Framework triples, and a compiled dissertation.

The contribution is fourfold: a theoretical operating model for governed multimodal epilepsy intelligence; a methodological pipeline demonstrating real-EEG, explainable, reproducible analysis; a practical deployable artefact; and an honest account of the boundary between what is implemented and what remains specified. The work argues that the decisive gap in clinical AI is not algorithmic but operational, and that value accrues when models are embedded in owned, governed, monitored pipelines with human oversight throughout.

Keywords: epilepsy, electroencephalography, explainable AI, multimodal fusion, MLOps, responsible AI, knowledge graph, retrieval-augmented generation, design science, clinical decision support.

Declaration

I declare that this dissertation is my own work and that all sources have been acknowledged in accordance with APA 7th-edition conventions. The empirical EEG analyses use publicly available, de-identified datasets (CHB-MIT via PhysioNet; EEG-Eye-State via OpenML). Clinical cohort data used for the severity and recurrence demonstrations are synthetic and are labelled as such throughout. The platform is a research and decision-support artefact; it is not a certified medical device and does not provide autonomous diagnosis.

Acknowledgements

The author acknowledges the PhysioNet and OpenML communities for open datasets, the maintainers of the open-source scientific Python and JavaScript ecosystems on which the artefact is built, and the clinical framing provided by the International League Against Epilepsy classification.

Table of Contents

Caption — the dissertation structure; each chapter carries a flowchart, sequence diagram, network

diagram, and C4 model, with supporting tables and figures.*

Chapter	Title
5	Implementation
6	Results & Evaluation
7	Discussion
8	Conclusion, Limitations & Future Work
References	Consolidated APA-7 references (per chapter)
Appendices	Governance pack, enterprise-flow specifications, knowledge-graph ontology

List of Abbreviations

Caption — abbreviations used throughout the dissertation.

Abbrev.	Meaning
AI	Artificial Intelligence
ASM	Anti-Seizure Medication
AUC	Area Under the ROC Curve
CWT	Continuous Wavelet Transform
DBA	Doctor of Business Administration
DSR	Design Science Research
EEG	Electroencephalography
HIPAA	Health Insurance Portability and Accountability Act
ILAE	International League Against Epilepsy
KG	Knowledge Graph
MCP	Model Context Protocol
MLOps	Machine-Learning Operations
NIST	National Institute of Standards and Technology
PLV	Phase-Locking Value
PR-AUC	Area Under the Precision-Recall Curve
RAG	Retrieval-Augmented Generation
RDF	Resource Description Framework
ROC	Receiver Operating Characteristic
SHAP	SHapley Additive exPlanations
SMOTE	Synthetic Minority Over-sampling Technique
STFT	Short-Time Fourier Transform

List of Figures

Caption — the 30 figures; every chapter carries a flowchart, sequence diagram, network diagram, and C4 model.

List of Tables

Caption — the 19 numbered tables across the dissertation.

Thesis Structure Map

The dissertation comprises **8 chapters, 73 numbered sections, 30 figures, and 19 tables**. The structure is presented below as a logical-sequence flowchart, a chapter table, a bullet outline, a dependency network, and a proportion chart.

Structure as a table

Caption — chapters with their section, figure, and table counts.

Ch	Title	Sections	Figures	Tables
1	Introduction	7	4	2
2	Literature Review	13	4	2
3	Research Methodology	9	4	3

Ch	Title	Sections	Figures	Tables
4	System Design & Architecture	12	5	3
5	Implementation	11	4	3
6	Results & Evaluation	10	5	4
7	Discussion	6	2	0
8	Conclusion, Limitations & Future Work	5	2	2
	Total	**73**	**30**	**19**

Structure as a logical-sequence flowchart

Figure FM.1 — the dissertation's logical reading sequence.

■ *Diagram (Mermaid — interactive in the viewer): flowchart TD*

Structure as a bullet outline

- **Front matter** — Abstract · Declaration · Acknowledgements · Table of Contents · List of Figures · List of Tables · List of Abbreviations
- **Chapter 1 — Introduction:** 1.1 Background · 1.2 Business & Clinical Problem · 1.3 Aim, Questions, Hypotheses · 1.4 Scope · 1.5 Platform in Outline · 1.6 Significance & Contribution · 1.7 Structure
- **Chapter 2 — Literature Review:** 2.1 Scope · 2.2 Review Method · 2.3 Epilepsy & Diagnosis · 2.4 ML for Seizure Detection (2.4.1 Classical · 2.4.2 Deep Learning · 2.4.3 Benchmarks) · 2.5 Time-Frequency & Images · 2.6 Multimodal Fusion · 2.7 Explainable AI · 2.8 Responsible AI & Standards · 2.9 MLOps · 2.10 RAG & Knowledge Graphs · 2.11 Synthesis & Gap · 2.12 Delimitation · 2.13 Summary
- **Chapter 3 — Methodology:** 3.1 Orientation · 3.2 DSR Cycle · 3.3 Questions & Hypotheses · 3.4 Unit/Severity/Roles · 3.5 Data Sources · 3.6 Collection & Analysis Interaction · 3.7 Analytical Methods · 3.8 Rigor, Validity, Ethics · 3.9 Summary
- **Chapter 4 — System Design:** 4.1 Rationale · 4.2 Seven Pipelines · 4.3 Forty-Stage Architecture · 4.4 Data Architecture · 4.5 Feature Architecture · 4.6 Serving · 4.7 Generative-AI · 4.8 Governance-by-Design · 4.9 Prediction Object · 4.10 C4 Views · 4.11 Maturity · 4.12 Summary
- **Chapter 5 — Implementation:** 5.1 Philosophy · 5.2 Signal Processing (real EEG) · 5.3 Time-Frequency/CV · 5.4 Feature Engineering · 5.5 Imbalance & Modelling · 5.6 Primary/Fusion (synthetic) · 5.7 MLOps · 5.8 GenAI · 5.9 Component Model · 5.10 Tech Stack · 5.11 Reproducibility & Viewer
- **Chapter 6 — Results & Evaluation:** 6.1 Philosophy · 6.2 Protocol · 6.3 Real-EEG Detection (6.3.1 External Validation · 6.3.2 Model Comparison) · 6.4 Significance/Ranking/Ablation · 6.5 Explainability · 6.6 Synthetic Clinical/Fusion · 6.7 Fairness · 6.8 Operational/Governance · 6.9 Hypothesis Outcomes · 6.10 Threats to Validity
- **Chapter 7 — Discussion:** 7.1 Overview · 7.2 Interpreting Performance · 7.3 Seven Pipelines · 7.4 Explainability/RAG/KG · 7.5 Governance & Trust · 7.6 Organisational/DBA Implications
- **Chapter 8 — Conclusion:** 8.1 Problem & Response · 8.2 Contributions · 8.3 Limitations · 8.4 Future Work · 8.5 Reflective Closing

Structure as a dependency network

Figure FM.2 — how chapters depend on one another beyond linear order.

■ *Diagram (Mermaid — interactive in the viewer): graph LR*

Structure as a chart

Figure FM.3 — proportion of numbered sections per chapter.

■ *Diagram (Mermaid — interactive in the viewer): pie showData*

Chapter 1 — Introduction

1.1 Background

Epilepsy is among the most common serious neurological disorders, affecting an estimated fifty million people worldwide and imposing a substantial burden on patients, families, and health systems. It is characterised by a persisting predisposition to recurrent, unprovoked seizures, and its clinical management depends on the synthesis of two very different kinds of evidence: the narrative clinical history and examination recorded by a neurologist, and the electrophysiological signal recorded by an electroencephalography technician. In routine practice these two evidence streams are gathered by different professionals, stored in different systems, and interpreted at different times. The consequence is a diagnostic pathway that is frequently slow, expensive, fragmented, and reactive rather than anticipatory. Patients may wait weeks for an assessment that integrates their history with their EEG, and clinicians must reconstruct a longitudinal picture from records that were never designed to be read together.

Artificial intelligence has been proposed repeatedly as a remedy, and the research literature is rich with models that detect or predict seizures from EEG with impressive accuracy on benchmark datasets. Yet the translation of these models into dependable clinical operation has lagged far behind their reported performance. A model is not a service. A healthcare organisation that wishes to use a seizure detector must first solve a long chain of problems that the typical research paper does not address: how the data arrives and is contracted, how it is governed and versioned, how features are computed without leakage, how the model is registered, served, monitored for drift, and rolled back, how its outputs are explained and audited, and how patient consent and privacy are enforced throughout. This dissertation takes the position that the decisive gap in clinical epilepsy AI is therefore not primarily algorithmic but **operational and organisational**, which is precisely the domain of a Doctor of Business Administration inquiry.

1.2 The Business and Clinical Problem

The organisational problem motivating this work can be stated plainly. Current epilepsy assessment consumes scarce specialist time, produces fragmented records, and recognises clinical deterioration late. The clinical problem that follows is to support — never to replace — the neurologist and EEG technician by fusing their evidence, estimating seizure severity and near-term risk, explaining every recommendation, and doing so within a governed workflow that a health system can trust and audit. The research problem, in turn, is to design and evaluate an enterprise-grade platform that delivers this support reproducibly, on real physiological data, with explicit responsible-AI and safety controls.

Caption — Table 1.1 traces the line from the operational pain point to the research response,

establishing that the contribution is an operating model, not a single algorithm.*

Layer	Statement
Business problem	Specialist assessment is slow, costly, fragmented; deterioration recognised late
Business objective	Improve a measurable clinical/operational outcome via an AI-supported process
Clinical problem	Fuse clinical + EEG evidence; estimate severity and 90-day seizure risk; explain and govern
Research problem	Design + evaluate an enterprise, explainable, multimodal, governed epilepsy platform
Research objective	Deliver organisational value across clinical, technical, operational, and governance dimensions

1.3 Research Aim, Questions, and Hypotheses

The aim of the dissertation is to design, implement, and evaluate an explainable multimodal AI platform for epilepsy that demonstrably embeds machine learning within a governed enterprise operating model. Five research questions structure the inquiry, and each is paired with a testable hypothesis evaluated in Chapter 6.

Caption — Table 1.2 maps each research question to its hypothesis and the principal source of

evidence, foreshadowing the honest separation between real-EEG findings and synthetic demonstrations.*

#	Research question	Hypothesis (H1)	Evidence
RQ1	Can interpretable EEG biomarkers discriminate ictal states?	They discriminate above chance	Real CHB-MIT EEG
RQ2	Does the detector generalise beyond its training recording?	It generalises to an external benchmark	EEG-Eye-State external validation
RQ3	Does explainability identify clinically legible drivers?	Top drivers are clinically meaningful	SHAP / permutation importance
RQ4	Is governance-by-design (monitoring, safety, consent) feasible?	It is feasible and auditable	Phase gates, governance pack
RQ5	Does an enterprise operating model improve deployability?	Seven owned pipelines improve maturity	40-stage architecture status

1.4 Scope and Delimitations

The scope is deliberately bounded to epilepsy. The empirical claims that matter are grounded in real scalp EEG; the clinical severity, recurrence, and fusion models are demonstrated on a synthetic cohort and are labelled as such wherever they appear, because responsible practice forbids presenting synthetic performance as clinical validation. The platform is a research and decision-support artefact, not a certified medical device, and every predictive output is subject to neurologist review. Deep neural architectures are represented by a lightweight multilayer-perceptron stand-in, and the retrieval-augmented-generation layer uses lexical embeddings; these are explicit scoping choices whose production replacements are identified in Chapter 8.

1.5 The Platform in Outline

The platform is organised as seven connected pipelines, each with a distinct owner and control set, rather than as a single fused flow. Figure 1.1 shows the end-to-end progression from research protocol to governed clinical output, with the feedback loop through which monitoring and new labels return to the modelling stages.

Figure 1.1 — Flowchart of the seven-pipeline operating model and its feedback loop.

■ *Diagram (Mermaid — interactive in the viewer): flowchart TD*

The interaction between the human actors and the platform is equally important. Figure 1.2 shows, at the level of a single encounter, how a clinician's request is authenticated, grounded in features, scored, explained, safety-checked, and returned for human approval, with every step audited.

Figure 1.2 — Sequence diagram of a governed, audited prediction within one clinical encounter.

■ *Diagram (Mermaid — interactive in the viewer): sequenceDiagram*

The platform's components and their data relationships are summarised in Figure 1.3, which situates the data sources, the modelling and serving core, and the responsible-AI and knowledge layers.

Figure 1.3 — Network diagram of the platform's principal components and data flows.

■ *Diagram (Mermaid — interactive in the viewer): graph LR*

Finally, Figure 1.4 gives a Context-level C4 view, naming the human roles, the platform as a single system, and the external systems on which it depends.

Figure 1.4 — C4 context model situating the platform among its users and external systems.

■ *Diagram (Mermaid — interactive in the viewer): graph TB*

1.6 Significance and Contribution

The significance of the work is that it reframes clinical epilepsy AI as an enterprise operating problem and demonstrates, on real data, that the pieces can be assembled coherently and honestly. Its contribution is theoretical, in offering a governed multimodal operating model; methodological, in providing a reproducible, explainable, real-EEG pipeline; practical, in delivering a runnable artefact with an interactive viewer and a compiled dissertation; and reflective, in drawing a candid boundary between the implemented and the specified. For a DBA, the emphasis falls on organisational value — ownership, governance, adoption, and risk — rather than on algorithmic novelty alone.

1.7 Structure of the Dissertation

Chapter 2 reviews the literature on EEG machine learning, multimodal fusion, explainable and responsible AI, MLOps, and knowledge-grounded generation, and locates the gap this work fills. Chapter 3 sets out the Design Science methodology, data sources, variables, hypotheses, and ethics. Chapter 4 presents the system design and architecture, and Chapter 5 the implementation on real EEG and the supporting engineering. Chapter 6 reports the results and evaluation against the five hypotheses. Chapter 7 discusses the findings and their organisational implications, and Chapter 8 concludes with contributions, a frank account of limitations, and a future-work agenda.

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Chapter 2 — Literature Review

2.1 Introduction and Scope of the Review

This chapter critically synthesises the body of scholarship that informs the design, implementation, and evaluation of an enterprise-grade, explainable, multimodal artificial intelligence (AI) platform for remote epilepsy care. Rather than cataloguing prior work in isolation, the review is organised around the argument that although individual technical components of such a platform are well studied, the literature contains a persistent integration gap: prior research overwhelmingly optimises isolated models for narrowly defined tasks and rarely delivers a governed, deployable, end-to-end system that spans the full data-to-model-to-generative-AI-to-governance lifecycle for epilepsy. Establishing this gap requires traversing several distinct literatures — the clinical epidemiology and diagnostic pathway of epilepsy, signal processing and machine learning for electroencephalography (EEG), explainable and responsible AI, machine learning operations (MLOps) and clinical prediction-model reporting standards, and retrieval-augmented generation (RAG) with knowledge graphs for clinical decision support. Each literature is mature in its own right, yet the seams between them remain thinly theorised and rarely engineered.

The review deliberately restricts its clinical scope to epilepsy, the fourth most common neurological disorder worldwide, and treats scalp EEG as the primary physiological modality of interest. It proceeds by first grounding the reader in the clinical problem, then moving through the computational methods that have been applied to it, and finally examining the organisational, ethical, and engineering literatures that determine whether such methods ever reach patients. The chapter closes by consolidating the identified gaps into a synthesis matrix and situating the present study within the field.

2.2 Review Methodology and Corpus Selection

The corpus underpinning this review was assembled through a structured search of PubMed, IEEE Xplore, the ACM Digital Library, Scopus, and Google Scholar, supplemented by backward and forward citation chasing from seminal works. Search terms combined clinical vocabulary (epilepsy, seizure, EEG, ILAE classification) with computational vocabulary (machine learning, deep learning, seizure detection, seizure prediction, explainable AI, MLOps, retrieval-augmented generation). Inclusion favoured peer-reviewed articles, foundational conference papers, and authoritative standards documents; exclusion removed non-English items, works addressing neurological conditions other than epilepsy without transferable methodology, and grey-literature marketing material. Figure 2.1 depicts the PRISMA-style selection flow used to arrive at the reviewed corpus.

■ *Diagram (Mermaid — interactive in the viewer): flowchart TD*

Figure 2.1. PRISMA-style flow of literature identification, screening, and inclusion for the review corpus.

2.3 Epilepsy: Burden, Classification, and the Diagnostic Pathway

Epilepsy affects an estimated fifty million people globally and carries a burden that extends well beyond seizure events to encompass cognitive comorbidity, psychosocial stigma, injury risk, and premature mortality, including sudden unexpected death in epilepsy (World Health Organization, 2019). The diagnostic pathway remains clinically demanding: it depends on careful history taking, eyewitness accounts of episodes, and neurophysiological investigation, all of which are subject to interobserver variability and access inequities that are especially acute in remote and low-resource settings. The International League Against Epilepsy (ILAE) substantially revised the conceptual and operational framework for the disorder through its 2017 classifications, which reorganised seizures by onset (focal, generalised, or unknown) and awareness, and defined epilepsy as a disease of the brain rather than a single event (Fisher et al., 2017; Scheffer et al., 2017). This classificatory scaffolding is consequential for computational work because it defines the label space that any detection or prediction model must respect; a model trained without regard to onset type or awareness collapses clinically distinct phenomena into a single target and thereby limits its interpretive value.

Scalp EEG remains the cornerstone investigation, recording the summed postsynaptic potentials of cortical pyramidal neurons across standardised electrode montages such as the international 10–20 system. Interpreting these recordings requires recognition of interictal epileptiform discharges, ictal rhythms, and their spatial and temporal evolution — a skill concentrated among a limited pool of specialist neurophysiologists. The scarcity of this expertise, combined with the labour intensity of reviewing continuous multi-day recordings, is precisely the pressure that has motivated automated analysis. Yet the clinical literature is consistent in cautioning that EEG is a supportive rather than definitive test, that a normal interictal recording does not exclude epilepsy, and that overinterpretation is a recognised source of misdiagnosis. Any computational system therefore inherits an obligation not merely to classify but to explain, so that clinicians retain the epistemic authority the diagnosis demands.

2.4 Machine Learning for Seizure Detection and Prediction

2.4.1 Classical Feature Engineering

The first generation of automated EEG analysis was built on handcrafted features grounded in signal-processing theory. Spectral band power across the delta, theta, alpha, beta, and gamma bands captures the redistribution of rhythmic energy that accompanies ictal transitions. Line-length, introduced as a computationally efficient measure sensitive to both amplitude and frequency change, became a widely adopted marker for seizure onset because of its favourable trade-off between detection sensitivity and real-time tractability (Esteller et al., 2001). Hjorth parameters (activity, mobility, and complexity) summarise signal dynamics in the time domain, while entropy measures — sample entropy, approximate entropy, and spectral entropy — quantify the loss of complexity frequently observed as the brain transitions toward a seizure. Connectivity and nonlinear descriptors such as phase-locking value and fractal dimension extend the feature space to capture inter-channel synchronisation and self-similarity, phenomena that accumulating evidence links to the mechanics of ictogenesis. The enduring appeal of these features is their interpretability: a clinician can reason about elevated line-length or diminished entropy in a way that resists the opacity of learned representations. Their limitation is equally clear, in that feature selection encodes strong prior assumptions and may fail to generalise across the heterogeneity of seizure semiology.

2.4.2 Deep Learning Approaches

Deep learning reframed the problem by learning representations directly from data. Convolutional neural networks (CNNs) applied to raw or minimally processed EEG can learn spatial and temporal filters that subsume many handcrafted features, and compact architectures designed specifically for EEG — most notably EEGNet, which combines depthwise and separable convolutions to remain parameter-efficient — have demonstrated competitive performance across brain-computer interface and clinical paradigms (Lawhern et al., 2018). Recurrent architectures and, more recently, transformer models that exploit self-attention have been applied to capture long-range temporal dependencies in continuous recordings. The published performance of these models is frequently impressive on held-out segments of benchmark datasets, yet the literature exhibits a recurring methodological fragility: results are often reported without patient-independent validation, without calibration analysis, and without external testing on data from other institutions or recording conditions. The consequence is a body of work that may overstate clinical readiness, a concern that connects directly to the reporting-standards literature discussed in Section 2.8.

2.4.3 Benchmarks and Their Discontents

Progress in the field has been anchored by shared benchmarks, foremost among them the CHB-MIT Scalp EEG Database, which provides continuous recordings from paediatric subjects with intractable seizures and has become a de facto standard for evaluating detection and prediction algorithms (Shoeb, 2009). Other resources, including the Temple University Hospital EEG Corpus, the Bonn University dataset, and the Kaggle seizure-prediction challenges, have broadened the evaluation landscape. Benchmarks have been indispensable for comparability, but they also impose a distorting gravity: models are tuned to idiosyncrasies of specific datasets, and reported gains may reflect dataset overfitting rather than genuine clinical robustness. Few studies pair a benchmark result with an external validation set drawn from a genuinely different distribution, and fewer still interrogate whether a model's decision rests on physiologically plausible signal rather than artefact. The present study responds to this critique by performing real CHB-MIT scalp EEG analysis for seizure detection while explicitly conducting external validation on the independent EEG-Eye-State dataset, thereby treating distribution shift as a first-class evaluation concern rather than an afterthought.

2.5 Time-Frequency Analysis and Image Representations of EEG

Because EEG is non-stationary, purely time-domain or purely frequency-domain descriptions are incomplete, and time-frequency methods have become central to both feature engineering and deep learning pipelines. The short-time Fourier transform (STFT) provides a fixed-resolution spectrogram, while the continuous wavelet transform (CWT) offers multiresolution analysis better suited to transient epileptiform events, trading time and frequency resolution adaptively across scales. A significant methodological current transforms one-dimensional EEG into two-dimensional images — spectrograms, scalograms, or recurrence plots — so that mature convolutional architectures developed for computer vision can be applied directly. This 1D-to-2D reformulation has proven productive, but it introduces its own interpretive hazards: the choice of window, mother wavelet, and colour mapping shapes the representation in ways that can either reveal or obscure the underlying physiology, and a model that appears to detect seizures may in fact be keying on representational artefacts. The literature would benefit from more systematic study of how these preprocessing choices interact with downstream explainability, an interaction that few papers examine and that the present platform addresses by pairing its representations with post hoc attribution analysis.

2.6 Multimodal Fusion of Clinical and Physiological Data

Seizures are rarely diagnosed from EEG alone; clinicians integrate history, medication response, comorbidity, imaging, and semiological description. Multimodal machine learning seeks to replicate and augment this integration by fusing heterogeneous data streams. The literature distinguishes early fusion, in which raw or feature-level representations are concatenated before modelling; late fusion, in which modality-specific models are combined at the decision level; and intermediate or hybrid strategies that learn joint representations (Baltrušaitis et al., 2019). Each strategy carries characteristic failure modes: early fusion is vulnerable to modality imbalance and missingness, late fusion can discard cross-modal interactions, and learned joint embeddings risk one dominant modality overwhelming weaker signals. In epilepsy specifically, principled fusion of continuous EEG with structured clinical variables remains comparatively underdeveloped, and studies frequently treat the clinical context as static metadata rather than as an information source with its own temporal dynamics. Figure 2.2 contrasts the linear data flow of a typical prior detection system with the richer, governed flow that a multimodal platform requires, foreshadowing the architectural argument of this dissertation.

■ *Diagram (Mermaid — interactive in the viewer): sequenceDiagram*

Figure 2.2. Sequence of data movement in a representative prior seizure-detection system, illustrating the unimodal, unexplained, and ungoverned flow this study seeks to remedy.

2.7 Explainable AI in Clinical Machine Learning

The clinical adoption of any predictive model is contingent on trust, and trust in high-stakes medicine is inseparable from explanation. Two model-agnostic techniques dominate the applied literature. SHAP (SHapley Additive exPlanations) grounds feature attribution in cooperative game theory, distributing a prediction across input features according to their marginal contributions and offering a unified framework that subsumes several earlier attribution methods (Lundberg & Lee, 2017). LIME (Local Interpretable Model-agnostic Explanations) instead fits a sparse, interpretable surrogate model in the local neighbourhood of a specific prediction, providing an intuitive account of which inputs drove a particular decision (Ribeiro et al., 2016). For neural models operating on signals or images, gradient-based saliency and class-activation mapping methods highlight the temporal or spatial regions most influential to the output. Complementing these, permutation importance offers a simple, model-agnostic global measure of feature relevance by observing the degradation in performance when a feature is randomised.

A substantial critical literature tempers enthusiasm for these tools. Attribution methods can be unstable, sensitive to hyperparameters and baselines, and vulnerable to adversarial manipulation, and there is a well-argued distinction between explanations that are faithful to a model's actual computation and those that are merely plausible to a human reader (Rudin, 2019). Rudin's argument that high-stakes decisions may warrant inherently interpretable models rather than post hoc explanation of black boxes remains a productive provocation for clinical AI, and it is one this study takes seriously by combining lightweight, tractable models with SHAP and permutation-based attribution so that explanation and computation remain closely coupled rather than divorced. The broader point is that explainability in epilepsy care is not a cosmetic add-on but a clinical safety requirement: a neurophysiologist must be able to see that a model's seizure call rests on physiologically meaningful signal features rather than on montage artefacts or dataset shortcuts.

2.8 Responsible AI, Governance, and Reporting Standards

Explanation addresses one facet of trustworthiness; fairness, security, and rigorous reporting address the rest. The National Institute of Standards and Technology AI Risk Management Framework provides a structured, function-oriented approach — govern, map, measure, and manage — for identifying and mitigating AI risks across the lifecycle, and it has rapidly become a reference point for organisations seeking to operationalise responsible AI (National Institute of Standards and Technology, 2023). Fairness scholarship has demonstrated both the plurality of formal fairness definitions and their mutual incompatibility, establishing that no single metric can satisfy all reasonable notions of equity simultaneously and that fairness choices are therefore normative rather than purely technical (Mehrabi et al., 2021). In epilepsy care, where access to specialist EEG interpretation is already inequitably distributed, an automated platform that performs unevenly across demographic or clinical subgroups risks entrenching rather than relieving disparity.

Parallel to these ethical frameworks runs a rigorous literature on the reporting and appraisal of clinical prediction models. The TRIPOD statement establishes reporting standards for the transparent development and validation of prediction models, insisting on clear description of data, predictors, model specification, and validation strategy (Collins et al., 2015). The PROBAST tool provides a structured means of assessing risk of bias and applicability in prediction-model studies, with particular attention to the analysis domain where optimism, overfitting, and inappropriate validation frequently arise (Moons et al., 2019). Read together, these instruments constitute a demanding standard against which much of the seizure-detection literature falls short, precisely because that literature so often omits external validation, calibration, and transparent reporting. The security dimension is captured by widely adopted engineering guidance, notably the OWASP application-security standards, and by the regulatory obligations of health-data protection regimes such as HIPAA. The present platform is designed explicitly against this triad — NIST for risk governance, OWASP for application security, and HIPAA for data protection — and is embedded within an institutional review board (IRB) approval and informed-consent framework, thereby treating governance as an architectural concern rather than a compliance afterthought.

2.9 MLOps and the Deployment of Clinical Machine Learning

Even a well-validated, fair, and explainable model delivers no clinical value until it is reliably deployed, monitored, and maintained, and here the literature on machine-learning operations becomes decisive. Sculley and colleagues' influential analysis of the hidden technical debt in machine-learning systems demonstrated that model code constitutes only a small fraction of a production system, which is dominated instead by data collection, feature management, serving infrastructure, configuration, and monitoring, and that neglect of these components accrues compounding maintenance costs (Sculley et al., 2015). This insight reframes the deployment challenge from a modelling problem to a systems-engineering problem. Subsequent MLOps scholarship has formalised concerns around data and concept drift — the tendency of input distributions and input-output relationships to change over time — and around the continuous monitoring, retraining, and versioning required to detect and remediate such drift before it degrades clinical performance. In continuous EEG monitoring, drift is not hypothetical: changes in acquisition hardware, electrode placement, patient population, and even seasonal or circadian factors can shift the signal distribution, and a static model silently decays. Robust data engineering, lineage tracking, and reproducibility are therefore not luxuries but preconditions for safe clinical operation. The present study operationalises these lessons through a seven-pipeline enterprise operating model that separates and instruments the stages of data ingestion, processing, modelling, explanation, generative augmentation, governance, and monitoring.

2.10 Retrieval-Augmented Generation and Knowledge Graphs in Clinical Decision Support

The most recent literature relevant to this platform concerns the integration of large language models into clinical decision support through retrieval-augmented generation. Lewis and colleagues introduced RAG as an architecture that couples a parametric generative model with a non-parametric retrieval component over an external corpus, allowing generated output to be grounded in retrieved evidence and reducing the propensity of language models to fabricate (Lewis et al., 2020). For clinical use, this grounding is essential, because unmoored generation is unacceptable where patient safety is at stake. Complementing retrieval over unstructured text, knowledge graphs offer a structured, machine-interpretable representation of clinical entities and their relations, and the maturing literature on knowledge-graph construction, representation, and reasoning provides the formal foundations for encoding epilepsy domain knowledge as interoperable triples (Hogan et al., 2021). The convergence of vector-based retrieval with graph-structured knowledge points toward decision-support systems that are simultaneously flexible and constrained by curated clinical facts. Yet the epilepsy-specific literature applying these techniques within a governed, explainable, and validated platform is sparse, and most demonstrations remain proofs of concept detached from the data-engineering and governance apparatus that clinical deployment demands. This platform bridges that divide by combining a RAG vector database with an RDF knowledge graph, situating generative augmentation inside the same governed lifecycle as its predictive models.

2.11 Thematic Synthesis and Identification of the Gap

The literatures reviewed above are individually deep but collectively disjointed. Figure 2.3 maps the principal thematic clusters and the sparse connective tissue between them, and Figure 2.4 situates the present research as a context model that integrates these clusters around the patient and the clinician. The recurring pattern across all seven clusters is one of local optimisation without systemic integration: EEG modelling advances without governance, explainability advances without deployment, and generative decision support advances without validation. What is largely absent is a single, coherent, enterprise-grade platform that carries a signal from raw acquisition through validated modelling, faithful explanation, grounded generative augmentation, and enforced governance, all while respecting the clinical realities of epilepsy care.

■ *Diagram (Mermaid — interactive in the viewer): graph LR*

Figure 2.3. Thematic network map of the reviewed literature clusters, illustrating strong intra-cluster development and comparatively weak inter-cluster integration.

■ *Diagram (Mermaid — interactive in the viewer): flowchart TB*

Figure 2.4. C4-style context model situating the present research among the patient, clinician, and external systems, integrating the otherwise disjoint literature clusters.

Table 2.1 presents a synthesis matrix contrasting representative strands of prior work with the present study across the dimensions that define an enterprise-grade platform.

Table 2.1. Synthesis matrix of representative prior work versus the present study.

Dimension	Classical EEG ML (e.g., Esteller et al., 2001)	Deep EEG models (e.g., Lawhern et al., 2018; Shoeb, 2009)	Clinical XAI (Lundberg & Lee, 2017; Ribeiro et al., 2016)	RAG / KG decision support (Lewis et al., 2020; Hogan et al., 2021)	This Study
Epilepsy-specific EEG modelling	Yes	Yes	Rare	Rare	Yes (real CHB-MIT)
External validation	Uncommon	Uncommon	N/A	N/A	Yes (EEG-Eye-State)
Explainability	Inherent but narrow	Limited/post hoc	Central	Limited	SHAP + permutation
Multimodal fusion	No	Occasional	No	Text-centric	Yes (clinical + EEG)
Governance (NIST/HIPAA/OWASP)	No	No	No	Rare	Yes
Generative decision support	No	No	No	Yes	RAG + RDF graph
Enterprise operating model	No	No	No	No	7-pipeline model
Ethical oversight (IRB/consent)	Varies	Varies	Varies	Varies	Yes

Table 2.2 distils these observations into an explicit gap analysis, naming each gap and the corresponding response embodied in the platform.

Table 2.2. Gap analysis linking literature deficiencies to the present study's response.

Identified gap in the literature	Consequence	Response in this study
Models validated only in-distribution	Overstated clinical readiness	External validation on an independent dataset
Explanation decoupled from computation	Plausible but unfaithful explanations	Lightweight models paired with SHAP and permutation attribution
Governance treated as compliance afterthought	Unsafe, non-auditable deployment	NIST, HIPAA, OWASP, and IRB embedded architecturally
Isolated models without lifecycle integration	Hidden technical debt, silent drift	Seven-pipeline enterprise operating model with monitoring
Generative support ungrounded and unvalidated	Fabrication risk in clinical use	RAG vector database grounded by an RDF knowledge graph
Unimodal focus	Loss of clinical context	Fusion of physiological and clinical data

2.12 Scoping Choices and Honest Delimitation

Intellectual honesty requires that the present study's own boundaries be stated as plainly as the gaps it addresses. The clinical cohort data used to exercise the multimodal and governance components of the platform are synthetic rather than drawn from real patients, and the deep-learning components are deliberately lightweight stand-ins rather than large-scale production networks. These are scoping decisions appropriate to a design-and-evaluation dissertation whose contribution is the integrated platform architecture and its governed lifecycle, not a new state-of-the-art detection score. The seizure-detection evidence rests on genuine CHB-MIT scalp EEG analysis with external validation on EEG-Eye-State, which anchors the modelling claims in real data, while the synthetic cohort and compact models allow the fusion, explainability, generative, and governance mechanisms to be demonstrated end to end without incurring the ethical and logistical burden of a full clinical dataset at this stage. Framed this way, these choices are not overclaims but explicit delimitations that a subsequent clinical validation study would relax. The literature reviewed in this chapter both motivates the platform and establishes the standards — TRIPOD, PROBAST, NIST, and the fairness and explainability critiques — against which such a future validation must be judged.

2.13 Chapter Summary

This chapter traversed seven interlocking literatures: the clinical foundations of epilepsy and its diagnostic pathway; classical and deep-learning approaches to seizure detection and prediction; time-frequency and image representations of EEG; multimodal fusion; explainable AI; responsible AI, governance, and reporting standards; MLOps and deployment; and retrieval-augmented generation with knowledge graphs. Across all of them, the dominant scholarly pattern is deep local development coupled with weak systemic integration. Prior work optimises isolated components — a better classifier, a sharper explanation, a more fluent generator — but rarely assembles these into a governed, explainable, multimodal, deployable epilepsy platform spanning the full data-to-model-to-generative-AI-to-governance lifecycle. That integration gap, made explicit in Tables 2.1 and 2.2 and situated in Figure 2.4, is the space this dissertation occupies. The chapters that follow translate this critical synthesis into a concrete design, implementation, and evaluation, returning throughout to the standards and cautions that the reviewed literature so consistently supplies.

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Chapter 3 — Research Methodology

3.1 Introduction and Philosophical Orientation

This chapter specifies the methodological architecture through which the central design problem of this dissertation is investigated: how to conceive, construct, and evaluate an enterprise-grade, explainable, multimodal artificial intelligence (AI) platform that supports remote epilepsy care without displacing clinical authority. The chapter articulates the philosophical stance adopted, the overarching research design, the research questions and their attendant hypotheses, the unit of analysis, the data sources, the analytical procedures, and the provisions made for rigor, validity, and ethics. Throughout, the emphasis is on epilepsy as a chronic neurological condition characterised by recurrent seizures, and on the specific instrumentation—scalp electroencephalography (EEG), structured clinical assessment, and role-differentiated decision support—required to reason about ictal and inter-ictal states at a distance.

The philosophical stance of this study is pragmatism. Pragmatism holds that the value of a knowledge claim is judged by its practical consequences and its capacity to resolve a concrete problem, rather than by allegiance to a single ontological or epistemological doctrine (Creswell & Plano Clark, 2018). This orientation is well matched to a dissertation whose object is an artefact—a working software platform—that must simultaneously satisfy statistical, clinical, operational, and ethical criteria. Pragmatism authorises a mixed-methods posture in which quantitative model evaluation coexists with qualitative appraisal of explainability and workflow fit, and it licenses the use of whichever method most credibly answers the question at hand. Because the deliverable is a designed artefact evaluated in use, the study is framed within Design Science Research (DSR) as formalised by Hevner, March, Park, and Ram (2004) and operationalised through the process model of Peffers, Tuunanen, Rothenberger, and Chatterjee (2007). DSR treats the construction and evaluation of purposeful artefacts as a legitimate mode of scholarly inquiry, insisting that design knowledge be produced through rigorous cycles that connect a real-world problem environment to an accumulated knowledge base.

3.2 Research Design: Design Science Research Cycle

The study is organised as a single, integrated DSR cycle comprising six activities drawn from Peffers et al. (2007): identification of the problem and motivation, definition of objectives for a solution, design and development, demonstration, evaluation, and communication. These activities are not strictly linear; the design incorporates feedback loops so that evaluation findings inform iterative refinement of the artefact before the results are communicated. The relevance cycle grounds the work in the epilepsy care environment, the rigor cycle draws on and contributes to the knowledge base of biomedical signal processing and explainable machine learning, and the central design cycle iterates between building and evaluating (Hevner, 2007). Figure 3.1 renders this design flow.

Figure 3.1. End-to-end Design Science Research design flow for the explainable multimodal epilepsy platform.

■ *Diagram (Mermaid — interactive in the viewer): flowchart TD*

The problem identification activity establishes that patients living with epilepsy in remote or under-served settings experience fragmented monitoring, delayed detection of deterioration, and limited access to specialist interpretation of EEG. The objectives activity translates this problem into four solution properties—multimodal fusion, human-interpretable explanation, governable operation, and demonstrable operational efficiency—each of which becomes a testable claim. Design and development produce the artefact: an EEG analysis pipeline, a synthetic clinical cohort, a suite of predictive models, an explainability layer, and role-specific portals. Demonstration exercises the artefact on the reference test patient EP001 and on the CHB-MIT record chb01_03. Evaluation applies the mixed-methods battery described in Section 3.7. Communication is discharged through this dissertation and through a one-command reproducible pipeline.

The dotted refinement and relevance loops in Figure 3.1 signify that the artefact was revised in response to evaluation and that its ultimate justification rests on relevance to the epilepsy care environment.

3.3 Research Questions and Hypotheses

Five research questions (RQ1–RQ5) structure the inquiry, each paired with a directional hypothesis (H1–H5), an explicit null (H0), and a pre-specified statistical test. RQ1 asks whether combining multiple data modalities improves diagnostic decision support relative to a single-modality baseline. RQ2 asks whether explainability artefacts improve clinician-rated trust and appropriate reliance. RQ3 asks whether quantitative EEG biomarkers discriminate ictal from inter-ictal states. RQ4 asks whether continuous governance and monitoring of the deployed models is technically feasible and sustains performance within pre-declared bounds. RQ5 asks whether the platform confers operational efficiency gains in the remote care workflow. Table 3.1 maps each question to its hypothesis pair and its confirmatory test.

Table 3.1. Mapping of research questions to hypotheses and statistical tests.

RQ	Research Question	H0 (Null)	H1 (Alternative)	Statistical Test
RQ1	Does multimodal fusion improve diagnostic support over the best single modality?	AUC_multimodal ≤ AUC_unimodal	AUC_multimodal > AUC_unimodal	DeLong test for two correlated ROC curves
RQ2	Does explainability improve clinician trust and appropriate reliance?	Median trust_XAI ≤ median trust_control	Median trust_XAI > median trust_control	Wilcoxon signed-rank / Mann–Whitney U
RQ3	Do EEG biomarkers discriminate ictal from inter-ictal states?	Distributions of feature X are equal across states	Distributions differ across states	Mann–Whitney U with Benjamini–Hochberg FDR
RQ4	Is continuous governance/monitoring feasible within performance bounds?	Post-deployment AUC drift ≥ δ threshold	Drift < δ (stable within bounds)	Population Stability Index; PSI/KS drift test
RQ5	Does the platform yield operational efficiency gains?	Median time-to-triage_platform ≥ baseline	Median time-to-triage_platform < baseline	Mann–Whitney U on task-completion time

The confirmatory logic is deliberately conservative. For RQ1, the DeLong test is appropriate because the multimodal and unimodal classifiers are evaluated on the same patients, producing correlated receiver operating characteristic (ROC) curves whose area-under-the-curve (AUC) difference must be assessed with a paired procedure (DeLong, DeLong, & Clarke-Pearson, 1988). For RQ3, because dozens of candidate biomarkers are screened, the Mann–Whitney U statistic is accompanied by Benjamini–Hochberg control of the false discovery rate to guard against inflated Type I error under multiple comparisons. For RQ4, distributional drift in incoming EEG feature distributions is quantified with the Population Stability Index and the Kolmogorov–Smirnov statistic, operationalising "feasibility" as the platform's ability to detect and flag drift before performance degrades beyond a declared tolerance. The alpha level is fixed at 0.05 throughout, and effect sizes (rank-biserial correlation for U tests, AUC differences with 95% confidence intervals for DeLong) are reported alongside p-values so that practical, not merely statistical, significance can be judged.

3.4 Unit of Analysis, Severity Model, and Clinical Roles

The primary unit of analysis is the patient-encounter: a bounded episode in which a patient's EEG segment and structured clinical assessment are jointly evaluated to yield a severity classification and a decision-support recommendation. For biomarker discrimination (RQ3), a secondary unit is the EEG epoch, a fixed-length window of multichannel signal. The reference test patient EP001 anchors the demonstration activity; EP001 is the canonical case through which every module of the platform is exercised end-to-end, ensuring that the same instantiated pathway is traceable from raw signal to role-specific output.

Severity is represented through a four-level model aligned with the classification framework of the International League Against Epilepsy (ILAE) (Fisher et al., 2017; Scheffer et al., 2017). Level 1 (Mild) denotes infrequent seizures with full inter-ictal function; Level 2 (Moderate) denotes recurrent seizures with partial functional impact; Level 3 (Severe) denotes frequent or poorly controlled seizures with substantial impairment; and Level 4 (Refractory/Status) denotes drug-resistant epilepsy or status epilepticus requiring escalation. Because these levels are ordered, ordinal regression is the natural modelling choice for severity, and the ordering is preserved in all evaluation metrics. The platform serves ten differentiated clinical roles—neurologist, EEG technician, nurse, neuropsychologist, pharmacist, caregiver, patient, administrator, occupational therapist, and radiologist—each receiving a portal view calibrated to its information needs and authority. The neurologist retains diagnostic authority; all other roles receive decision support scoped to their remit, reflecting the human-in-the-loop constraint elaborated in Section 3.8.

3.5 Data Sources

The study draws on both secondary and primary data, and the provenance of each is stated with complete honesty because the credibility of a DSR artefact rests on transparent grounding. The secondary data are real, publicly available physiological recordings. The primary clinical cohort is synthetic and is used to demonstrate the methodology pending institutional review board (IRB) approval for the collection of real clinical data. This distinction is maintained explicitly throughout the dissertation so that no synthetic result is mistaken for a validated clinical finding. Table 3.2 catalogues the sources.

Table 3.2. Data sources: secondary (real) versus primary (synthetic).

Source	Type	Nature	Specifics	Role in Study
CHB-MIT Scalp EEG Database (PhysioNet)	Secondary	Real	Record chb01_03; 23-channel scalp EEG; 256 Hz sampling; annotated seizure onset/offset	Primary EEG signal for ictal/inter-ictal discrimination (RQ3) and demonstration
EEG-Eye-State (OpenML, ID 1471)	Secondary	Real	14-channel Emotiv EEG; 14,980 instances; binary eye-state label	External validation of feature-extraction and classifier generalisation
Synthetic Clinical Cohort	Primary	Synthetic	500 patients (EP001–EP500); demographics, seizure history, severity labels, medication, comorbidity	Clinical fusion, severity modelling, fairness/subgroup analysis (RQ1, RQ2, RQ5)

The CHB-MIT Scalp EEG Database, hosted on PhysioNet, comprises recordings from paediatric subjects with intractable seizures; record chb01_03 contains an annotated seizure and is used as the demonstration signal because its onset and offset annotations permit supervised evaluation of ictal detection (Goldberger et al., 2000; Shoeb, 2009). The signal is sampled at 256 Hz across 23 channels arranged according to the international 10–20 system. The EEG-Eye-State dataset from OpenML furnishes an independent EEG corpus recorded under different hardware and conditions; it is employed not as a clinical epilepsy source but as an external stress test of whether the feature-extraction and modelling code generalises beyond the training corpus, thereby strengthening the external-validity argument. The synthetic clinical cohort of 500 patients, indexed EP001 through EP500, is generated to be clinically plausible—its marginal distributions of age, seizure frequency, antiseizure medication regimen, and comorbidity are drawn from ranges reported in the epidemiological literature—but it contains no real patient data. Its purpose is strictly methodological: to demonstrate the fusion, severity-classification, and fairness pipelines on a dataset of realistic dimensionality while the study awaits IRB-approved access to genuine clinical records. Table 3.3 formalises the variable dictionary that operationalises these sources.

Table 3.3. Variable dictionary (independent, dependent, and covariate variables).

Variable	Role	Type	Source
Spectral band power (δ, θ, α, β, γ)	Independent	Continuous	CHB-MIT EEG
Hjorth parameters (activity, mobility, complexity)	Independent	Continuous	CHB-MIT EEG
Higuchi fractal dimension	Independent	Continuous	CHB-MIT EEG
Spectral entropy	Independent	Continuous	CHB-MIT EEG
Phase-locking value (PLV)	Independent	Continuous	CHB-MIT EEG (channel pairs)
Line-length	Independent	Continuous	CHB-MIT EEG
Modality indicator (EEG-only vs multimodal)	Independent	Binary	Design factor
Explanation condition (SHAP vs none)	Independent	Binary	Design factor
Ictal-state label (ictal / inter-ictal)	Dependent	Binary	CHB-MIT annotations
Severity level (L1–L4)	Dependent	Ordinal	Synthetic cohort / clinical assessment
Clinician trust rating	Dependent	Ordinal (Likert)	Evaluation study
Time-to-triage	Dependent	Continuous	Workflow logs
Age, sex, seizure frequency	Covariate	Mixed	Synthetic cohort
Antiseizure medication class	Covariate	Categorical	Synthetic cohort

3.6 Data Collection and Analysis Interaction

The movement of data from raw acquisition to evaluated inference follows a fixed sequence designed to prevent leakage and to preserve auditability. Figure 3.2 depicts the interaction among the principal processing stages.

Figure 3.2. Sequence of data collection, preprocessing, feature extraction, modelling, and evaluation.

■ *Diagram (Mermaid — interactive in the viewer): sequenceDiagram*

The sequence begins with acquisition of raw EEG at 256 Hz together with the corresponding clinical records. Preprocessing applies a band-pass filter of 0.5–45 Hz to retain physiologically meaningful frequency content while suppressing baseline drift and high-frequency noise, a notch filter to remove power-line interference, and a re-referencing step to a common reference montage. The cleaned signal is epoched into fixed windows. Feature extraction computes the biomarker set enumerated in Table 3.3. Crucially, the subject-level split precedes model training: all epochs from a given subject are assigned wholly to either the training or the held-out partition, so that no subject contributes to both. This subject-level partitioning is the single most important leakage control, because epoch-level random splitting would allow temporally adjacent windows from the same recording to appear in both training and test sets, inflating performance through identity leakage. Class imbalance—ictal epochs being far rarer than inter-ictal ones—is addressed with the Synthetic Minority Over-sampling Technique (SMOTE), fitted exclusively inside each training fold; the held-out subjects are never resampled, preventing synthetic examples from contaminating evaluation. Hyperparameters are tuned with GridSearchCV nested within the training partition, and the fitted models emit predictions that flow to the evaluation stage, whose outputs include refinement signals such as drift indicators and subgroup performance gaps that feed back into the design cycle.

3.7 Analytical Methods and Variable Relationships

The analytical apparatus spans classical statistics, machine learning, survival analysis, and time-series modelling, each mapped to a specific inferential need. Discrimination of ictal states and severity classification employ logistic regression and ordinal regression as interpretable baselines, RandomForest as a nonlinear ensemble, and a multilayer perceptron (MLP) as a representation learner. Recurrence of seizures over follow-up is modelled with Cox proportional-hazards regression, which accommodates censored observations inherent in longitudinal epilepsy monitoring. Temporal patterns in seizure counts are modelled with a seasonal autoregressive integrated moving-average model with exogenous regressors (SARIMAX), permitting the incorporation of covariates such as medication adherence. Time-frequency representations are computed with the short-time Fourier transform (STFT) and the continuous wavelet transform (CWT), the latter being well suited to the non-stationary, transient morphology of epileptiform discharges. Explainability is delivered through SHapley Additive exPlanations (SHAP), which attribute each prediction to its constituent features on a principled game-theoretic basis (Lundberg & Lee, 2017), supplemented by permutation importance for global feature ranking. Fairness and subgroup analysis compare performance across demographic strata to detect disparate error rates. Figure 3.3 maps the variable relationships that these methods estimate.

Figure 3.3. Variables map linking independent variables to the dependent variable through mediators and confounders.

■ *Diagram (Mermaid — interactive in the viewer): graph LR*

The map clarifies the causal-theoretic assumptions under test. EEG biomarkers and modality act as independent variables directly influencing diagnostic support accuracy, the principal dependent variable. Explanation influences accuracy indirectly, through the mediator of clinician trust and appropriate reliance—the hypothesis being that explanations improve outcomes only insofar as they calibrate reliance rather than inducing automation bias. Severity level operates as an intermediate outcome that both depends on biomarkers and shapes downstream decision support. Age, sex, seizure frequency, and medication class are treated as confounders whose influence is adjusted for statistically and probed in the fairness analysis. Figure 3.4 situates these analytical modules within the C4-style container view of the research apparatus.

Figure 3.4. C4-style container model of the research apparatus (data sources, analysis modules, evaluation).

■ *Diagram (Mermaid — interactive in the viewer): flowchart TB*

The container model exposes three layers. The data-source layer holds the two real EEG corpora and the synthetic clinical cohort. The analysis layer transforms raw signal into features, fits the model suite, and generates SHAP attributions. The evaluation-and-governance layer computes discrimination and calibration metrics, executes the confirmatory statistical tests, and runs fairness and drift monitoring. The dotted feedback edge labelled "run_all, seed=42" signifies that the entire apparatus is orchestrated by a single reproducible entry point, closing the loop from evaluation back to data ingestion.

3.8 Rigor, Validity, and Ethics

Methodological rigor is pursued along the dimensions of reproducibility, internal validity, external validity, and leakage control. Reproducibility is enforced through a fixed random seed (seed = 42) applied to every stochastic operation—data splitting, SMOTE resampling, model initialisation, and hyperparameter search—and through a one-command run_all orchestrator that regenerates every table and figure from raw inputs, so that any reviewer can reproduce the reported results deterministically. Internal validity is protected by the subject-level splitting and in-fold-only SMOTE described in Section 3.6, by nested cross-validation that separates hyperparameter selection from performance estimation, and by calibrated probability outputs assessed via reliability curves and log-loss. External validity is examined by evaluating the trained feature-extraction and classification code on the independent EEG-Eye-State corpus, which differs in hardware, population, and task, thereby testing whether the pipeline generalises beyond its development data. The predictive-modelling components are documented and appraised in alignment with the Transparent Reporting of a multivariable prediction model for Individual Prognosis Or Diagnosis (TRIPOD) statement and the Prediction model Risk Of Bias ASsessment Tool (PROBAST), so that the study's reporting completeness and risk-of-bias profile are explicit and auditable (Collins, Reitsma, Altman, & Moons, 2015; Wolff et al., 2019).

Ethical governance is treated as constitutive of the design rather than as an afterthought. The study will proceed to collection of real clinical data only after IRB approval; until then, all clinical demonstration is conducted on the synthetic cohort. When real data are collected, informed consent will be obtained and governed by an end-user licence agreement (EULA), and all identifiers will be removed in accordance with the de-identification provisions of the Health Insurance Portability and Accountability Act (HIPAA). Most fundamentally, the platform is positioned as decision support, not autonomous diagnosis: the neurologist remains in the loop as the accountable decision-maker, and no module issues a definitive diagnosis without human adjudication. This human-in-the-loop constraint is the ethical keystone of the design, reconciling the pursuit of algorithmic capability with the professional and legal responsibilities that attend the care of people living with epilepsy.

3.9 Chapter Summary

This chapter has set out a pragmatist, Design Science Research methodology combining artefact construction with a mixed-methods evaluation. It has articulated five research questions and their hypotheses, each with a stated null and a pre-registered statistical test; defined the patient-encounter and EEG-epoch as units of analysis anchored by reference patient EP001; specified a four-level ILAE-aligned severity model and ten clinical roles; and catalogued its data sources with candid distinction between real secondary EEG and synthetic primary clinical data. It has detailed the analytical pipeline—preprocessing, leakage-controlled splitting, biomarker extraction, time-frequency analysis, a multi-method model suite, and SHAP-based explanation—and the rigor, validity, and ethical safeguards that govern it. The chapter thereby establishes the methodological foundation on which the implementation and evaluation reported in the subsequent chapters rest.

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Chapter 4 — System Design & Architecture

4.1 Introduction and Design Rationale

This chapter presents the system design and architecture for the explainable, multimodal artificial intelligence (AI) platform that underpins the remote epilepsy care programme evaluated in this dissertation. The purpose of the chapter is to translate the clinical and organisational requirements established in the preceding chapters into a coherent enterprise architecture that is defensible on grounds of security, scalability, explainability, and auditability. The design follows the principles of design-science research articulated by Hevner, March, Park, and Ram (2004), in which a purposeful artefact is constructed and evaluated against a real organisational problem — in this instance, the fragmented, episodic, and often inequitable delivery of epilepsy care to patients who live at a distance from specialist epileptology services.

A central argument of this chapter is that clinical machine learning at enterprise scale cannot be delivered as a single monolithic model wrapped in an application. Sculley et al. (2015) demonstrated that the machine-learning code in a production system is a small fraction of the surrounding infrastructure, and that hidden technical debt accumulates precisely at the boundaries where data engineering, modelling, and operations meet without clear ownership. Accordingly, the platform is conceived not as one fused flow but as seven connected yet independently governed pipelines, each with a defined owner, a defined set of artefacts, and a defined contract with its neighbours. The remainder of the chapter develops this operating model, the underlying data and feature architectures, the serving and generative-AI subsystems, and the cross-cutting governance that binds them, before summarising the current maturity of each component honestly.

4.2 The Seven-Pipeline Operating Model

The naive approach to building a clinical prediction service is to construct a single flow that ingests data, trains a model, and returns a score. Such a design conflates concerns that have fundamentally different rates of change, risk profiles, and accountable owners. Research protocol design changes on the cadence of ethics review; data engineering changes when source systems change; model retraining changes when drift is detected; and the generative and agentic layer changes as clinical guidelines and language models evolve. Binding these into one flow produces the entanglement that Sculley et al. (2015) named "changing anything changes everything" (the CACE principle). The platform therefore decomposes the work into seven pipelines with explicit boundaries, mirroring the domain-oriented decomposition advocated in data mesh thinking (Dehghani, 2022).

■ *Diagram (Mermaid — interactive in the viewer): flowchart TD*

Figure 4.1. The seven-pipeline operating model with cross-pipeline feedback. Green nodes denote pipelines whose core is implemented; amber nodes denote pipelines that are partly specified design.

Figure 4.1 shows the seven pipelines arranged as a directed flow from research protocol (P1) through to clinical safety and monitoring (P7), together with the feedback loop that returns operational signal to the upstream pipelines. The feedback loop is the feature that distinguishes an enterprise operating model from a linear project plan: monitoring in P7 does not merely observe the system but actively triggers retraining in P4, raises data-contract breaches back to P2, and surfaces new clinical evidence and fairness concerns that reopen the research question in P1. Each pipeline owns its artefacts and exposes a contract to the next, so that a change within a pipeline — for example, a new feature transformation in P3 — is absorbed locally rather than propagating uncontrolled through the whole system. Table 4.1 sets out the ownership boundaries in full.

Table 4.1. *The seven pipelines with accountable owner, primary responsibility, and key artefacts.*

Pipeline	Accountable owner	Primary responsibility	Key artefacts
P1 Research Protocol	Principal investigator / clinical lead	Problem framing, cohort definition, endpoint selection, IRB/ethics approval	Protocol, cohort spec, consent forms, statistical analysis plan
P2 Data Engineering & Governance	Data platform engineering	Ingestion, lakehouse zones, master patient index, data contracts, lineage	Ingestion jobs, zone tables, MPI, catalog entries, data contracts
P3 Data Prep & Feature Engineering	Feature/analytics engineering	Quality validation, transformations, point-in-time-correct features	Quality rules, feature definitions, offline/online feature tables
P4 Statistical & ML Modelling	Data science / biostatistics	Model training, internal/external validation, probability calibration	Training pipelines, validated models, calibration curves, model cards
P5 MLOps & Deployment	ML platform / DevOps	Model registry, batch and real-time serving, rollback, champion-challenger	Registry, container images, serving API, deployment manifests
P6 RAG/LLM/Agent Engineering	Applied GenAI engineering	Retrieval, knowledge graph, MCP tool servers, governed orchestration	Vector index, RDF graph, MCP servers, agent policies
P7 Clinical Safety & Monitoring	Responsible-AI / clinical safety officer	Drift and performance monitoring, fairness, incident and audit management	Monitors, fairness reports, audit logs, incident register

4.3 The Forty-Stage Enterprise Architecture

Beneath the seven pipelines sits a more granular reference architecture of forty stages that trace a unit of work from an unframed clinical problem to a retained, auditable prediction. The forty stages are not a separate design but a decomposition of the seven pipelines, so that each stage belongs to exactly one pipeline and inherits its ownership. The stages proceed, in summary, from problem definition and cohort assembly (P1); through ingestion, landing in the lakehouse, master-patient-index resolution, and quality gating (P2); through feature engineering and registration in the feature store (P3); through model training, cross-validation, external validation, calibration, and registration (P4); through packaging, batch and real-time serving, and champion-challenger promotion (P5); through retrieval-augmented generation, knowledge-graph grounding, and human oversight (P6); and finally through monitoring, drift detection, fairness auditing, incident handling, and retention or archival (P7). Enumerating every stage is unnecessary here; the salient design property is that the forty stages form an unbroken, individually observable chain in which no transition between stages is unowned or unlogged. This granularity is what allows the platform to answer, for any given prediction, precisely which data, which features, which model version, and which human decision produced it.

4.4 Data Architecture

The data architecture adopts a lakehouse pattern, combining the low-cost, schema-flexible storage of a data lake with the transactional guarantees and governance of a warehouse. Kleppmann (2017) frames the central challenge of such systems as maintaining correctness and evolvability as data flows through successive representations; the lakehouse zone model addresses this by making each transformation explicit and by treating the earliest representation as immutable. Data enters a landing zone exactly as received, is copied to an immutable raw zone that is never mutated in place, and then progresses through validation, quarantine, silver, gold, feature, serving, and archive zones. Immutability of the raw zone is a deliberate governance decision: it guarantees that any downstream artefact can be reconstructed and that the provenance of every prediction is traceable to bytes that were never altered. Table 4.2 defines each zone.

Table 4.2. *Lakehouse zones, their purpose, mutability, and typical consumers.*

Zone	Purpose	Mutability	Typical consumer
Landing	First point of arrival, source-fidelity capture	Transient	Ingestion validators
Raw	Permanent, unaltered system of record	Immutable (append-only)	Reprocessing, audit
Validated	Records passing schema and data-contract checks	Append-only	Feature engineering
Quarantine	Records failing quality gates, held for remediation	Write-once, review	Data stewards
Silver	Cleaned, conformed, de-duplicated entities	Versioned	Analytics, features
Gold	Curated, business-level aggregates and cohorts	Versioned	Modelling, reporting
Feature	Point-in-time-correct feature values	Versioned, time-indexed	Feature store
Serving	Low-latency, denormalised read models	Materialised	Online API
Archive	Cold retention for compliance and reproducibility	Immutable, sealed	Legal, reproducibility

Three architectural concepts govern how data moves through these zones. Data contracts specify, for each source, the schema, semantics, quality expectations, and service levels to which the producer commits; a breach of contract routes records to quarantine and raises a signal back to P2, rather than silently corrupting downstream models. A master patient index (MPI) performs entity resolution so that records describing the same individual across electroencephalography (EEG) systems, seizure diaries, medication records, and wearable devices are reconciled to a single patient identity — an essential precondition for a longitudinal view of an epilepsy patient's condition. A metadata catalogue records schemas, ownership, and end-to-end lineage, so that the path from a raw EEG file to a served prediction is queryable.

It is important to distinguish three organising paradigms that are frequently conflated. Data mesh (Dehghani, 2022) is an organisational and ownership pattern in which domain teams publish curated data products; data fabric is an integration pattern that provides a unified access and semantic layer across heterogeneous stores; and the lakehouse is a storage-and-compute pattern. These are complementary rather than competing: the platform stores data in a lakehouse, exposes domain-owned data products in the spirit of data mesh, and integrates them through a fabric-like catalogue and semantic layer.

■ *Diagram (Mermaid — interactive in the viewer): graph LR*

Figure 4.2. Component and data-relationship diagram, tracing data from clinical sources through the lakehouse zones and feature plane to the model service, API, generative-AI layer, and monitoring.

Figure 4.2 renders these relationships as a network. The diagram makes visible the single most important leakage-prevention control in the architecture: served features flow only from the online feature store, which is populated by point-in-time-correct offline computations, and never directly from mutable operational sources.

4.5 Feature Architecture

The feature architecture separates offline and online feature stores that share a single set of feature definitions. The offline store computes and holds historical feature values for training and batch scoring; the online store holds the latest values for low-latency serving. The unifying discipline is point-in-time-correct retrieval: when a training example for a patient at time *t* is assembled, only feature values that were knowable at or before *t* are joined. This prevents temporal leakage — the subtle and common error in which a model is inadvertently trained on information from the future relative to the prediction it must make, producing optimistic validation results that collapse in production. By sharing definitions between the offline and online stores, the architecture also eliminates training–serving skew, in which a feature is computed one way in training and another way at inference.

4.6 Serving Architecture

The serving layer exposes models through both batch and real-time paths. Batch scoring runs against the gold and feature zones for population-level tasks such as cohort risk stratification, while a real-time application programming interface serves individual predictions. The real-time service is implemented in FastAPI and exposes a small, deliberately constrained surface: a `/predict` endpoint that returns a governed prediction object, and a `/metrics` endpoint that exposes operational and model-health telemetry for scraping by the monitoring subsystem. Every request is authenticated with an `X-API-Key` header, and unauthenticated requests are rejected before any feature retrieval or inference occurs.

Behind the API, a model registry holds versioned, validated models together with their model cards and calibration artefacts. Promotion follows a champion-challenger pattern: a new challenger model is served in shadow or to a limited cohort alongside the incumbent champion, its performance and fairness are compared on live traffic, and promotion or rollback is an explicit, logged, reversible operation. Rollback to a previous registered version is a first-class capability rather than an emergency improvisation.

■ *Diagram (Mermaid — interactive in the viewer): sequenceDiagram*

Figure 4.3. Sequence of an authenticated, audited prediction request across the pipelines, from clinician request through authorisation, feature retrieval, inference, calibration, explanation, safety review, human oversight, and audit.

Figure 4.3 traces a single prediction request end to end. The sequence encodes several non-negotiable controls: authorisation and consent are verified before any patient data is touched; calibration and uncertainty estimation sit between the raw model output and the clinician; a fairness and safety stage can route high-risk or low-confidence cases to human review; and an immutable audit record carrying a unique `audit_id` is written before the response is returned. No prediction reaches a clinician without traversing this full chain.

4.7 Generative-AI Architecture

The generative-AI layer augments the predictive models with retrieval, structured knowledge, tool use, and governed orchestration. Retrieval-augmented generation (RAG) grounds language-model outputs in an authoritative corpus of epilepsy guidelines and evidence held in a vector database, so that natural-language explanations and clinical summaries cite retrievable sources rather than relying on the parametric memory of the model. Complementing the vector store is a Resource Description Framework (RDF) knowledge graph encoding an epilepsy ontology whose principal classes are Patient, SeizureType, EEGFeature, antiseizure medication (ASM), Outcome, and Guideline, connected by relations such as *diagnosed-with*, *treated-by*, *contraindicated-for*, and *recommended-by*. Where the vector store supports fuzzy semantic recall, the knowledge graph supports precise, explainable reasoning over clinically meaningful relationships — for example, retrieving guideline-endorsed ASM options for a given seizure type while excluding contraindicated agents.

Tool access is mediated by Model Context Protocol (MCP) servers, which expose capabilities — feature lookups, guideline retrieval, knowledge-graph queries — as governed, discoverable tools with explicit input and output schemas. A multi-agent orchestration coordinates these tools to accomplish composite tasks, but the orchestration is deliberately constrained: it operates under an allow-list of tools, cannot mutate clinical records, and terminates at a mandatory human-approval gate before any recommendation is issued to a patient or clinician. This governed-autonomy posture reflects the guidance of the NIST AI Risk Management Framework (2023), which emphasises that automation in high-stakes settings must be bounded by human oversight and demonstrable accountability.

4.8 Governance-by-Design as a Cross-Cutting Concern

Governance is not a pipeline but a property that cuts across all seven. The platform aligns its controls with the HIPAA Security Rule for protected health information, the NIST AI Risk Management Framework (2023) for AI-specific risk, and the OWASP guidance for application and large-language-model security, including the OWASP Top 10 for LLM Applications (OWASP Foundation, 2023). Protected data is encrypted with AES-256 at rest and protected with TLS in transit. De-identification is applied before data is used for modelling, and re-identification keys are held under separate control. Every use of patient data is bound to documented consent and IRB approval, and every consequential action — data access, inference, override, promotion, and rollback — is written to an append-only audit log. Table 4.3 maps the principal quality attributes to the mechanisms that satisfy them.

Table 4.3. *Architectural quality attributes and the mechanisms by which each is achieved.*

Quality attribute	Mechanisms in the architecture	Governing reference
Security	AES-256 at rest, TLS in transit, X-API-Key authentication, secrets isolation, de-identification, MCP tool allow-listing	HIPAA; OWASP (2023); NIST (2023)
Scalability	Lakehouse separation of storage and compute, offline/online feature stores, batch + real-time serving, stateless API	Kleppmann (2017)
Explainability	Calibrated probabilities with uncertainty, per-prediction attributions, RAG citations, RDF knowledge-graph reasoning	Hevner et al. (2004); NIST (2023)
Auditability	Immutable raw zone, end-to-end lineage catalogue, append-only audit log with audit_id, model registry versioning	Kleppmann (2017); Sculley et al. (2015)

4.9 The Governed Prediction Output Object

The visible artefact of the entire architecture is the governed prediction object returned by `/predict`. Rather than a bare score, every prediction is a structured object that carries its own provenance and governance metadata: the source data and feature versions that produced it; a calibrated probability accompanied by an explicit uncertainty estimate; a human-readable explanation with feature attributions; a fairness flag indicating whether subgroup guardrails were satisfied; supporting evidence retrieved from the vector store and knowledge graph; the human-review status recording whether a clinician approved or overrode the result; and the `audit_id` linking the prediction to its immutable audit record. This object operationalises the principle that in clinical AI a prediction is not merely a number but an accountable claim, and it is the mechanism through which the abstract quality attributes of Table 4.3 become concrete and inspectable at the point of care.

4.10 The C4 Architectural Views

To communicate the architecture at two levels of abstraction, the design is expressed using the C4 model. The Context view situates the platform among its human actors and external systems, while the Container view decomposes the platform into its principal deployable and data components.

■ *Diagram (Mermaid — interactive in the viewer): C4Context*

Figure 4.4. C4 Context view — the platform, its human actors, and the external systems with which it exchanges data.

■ *Diagram (Mermaid — interactive in the viewer): C4Container*

Figure 4.5. C4 Container view — the principal deployable and data containers within the platform boundary and their runtime relationships.

Figures 4.4 and 4.5 provide the two mandated C4 perspectives. The Context view (Figure 4.4) establishes the platform's responsibilities and trust boundaries with clinicians, patients, the responsible-AI function, and external clinical systems. The Container view (Figure 4.5) decomposes the platform into the API gateway, lakehouse, feature store, model registry and service, RAG and knowledge-graph subsystem, MCP and agent runtime, and the audit-and-monitoring store, showing how a request flows across them.

4.11 Honest Assessment of Architectural Maturity

Intellectual honesty about maturity is itself a governance obligation, and the platform's components sit at different levels of realisation. The statistical and machine-learning modelling pipeline (P4), the batch and real-time serving with model registry and rollback (P5), the monitoring and audit subsystem (P7), the RAG vector database, and the RDF knowledge graph are implemented and operational within the evaluated system. In contrast, several components are specified designs — documented in detail and architecturally committed, but not yet fully running in production. These include the full nine-zone lakehouse with automated quarantine, the master patient index and entity-resolution service, the online feature store with production point-in-time retrieval, and the MCP tool servers together with the multi-agent runtime and its human-approval gate. Presenting these as complete would misrepresent the artefact; presenting them as designed rather than implemented locates the contribution correctly within the design-science tradition, in which a rigorously specified and partially instantiated architecture is a legitimate and evaluable research output (Hevner et al., 2004). Chapter 5 proceeds to describe the implementation of the components that are operational, and Chapter 7 returns to the specified-but-unbuilt components as a roadmap for maturation.

4.12 Summary

This chapter has presented an enterprise architecture for a remote epilepsy care AI platform organised as seven independently governed pipelines rather than a single fused flow, decomposed further into a forty-stage reference architecture in which every stage is owned and observable. It has detailed the lakehouse data architecture with its immutable raw zone and quality gates, the offline and online feature stores with point-in-time-correct retrieval, the batch and real-time serving layer with champion-challenger promotion, and the generative-AI layer combining RAG, an RDF epilepsy knowledge graph, MCP tool servers, and governed multi-agent orchestration under a human gate. Governance-by-design has been shown to cut across all pipelines through encryption, de-identification, consent, and immutable audit, and to crystallise in the governed prediction object returned at the point of care. The chapter closed with an honest appraisal of which components are implemented and which remain specified designs, positioning the work squarely within design-science research.

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Chapter 5 — Implementation

5.1 Overview and Implementation Philosophy

This chapter documents how the platform described in Chapters 3 and 4 was actually constructed, moving from architectural intent to running code. The guiding principle throughout implementation was *honest engineering*: every capability claimed by the platform is backed by an executable artefact that a reviewer can regenerate from a clean checkout, and every place where a lightweight component substitutes for a heavier production dependency is disclosed explicitly rather than concealed. This discipline matters in a doctoral context because reproducibility and transparency are themselves contributions; a demonstration that quietly relies on fabricated numbers, or that cannot be re-run, undermines the very governance argument the platform advances.

The implementation is organised around a single-command entry point, `analysis/run_all.py`, which executes every pipeline in dependency order under a fixed random seed of 42. A reviewer types `python analysis/run_all.py` and the system rebuilds the synthetic cohort, runs the flagship clinical and electroencephalographic (EEG) analyses, executes the governance gates, regenerates every figure and report, and refreshes the retrieval and knowledge-graph layers. The same command runs in continuous integration on every push and on a weekly schedule, so the artefacts committed to the repository are provably the artefacts the code produces. Two clearly separated data regimes underpin the platform. The secondary EEG pipeline operates on *real* scalp EEG — recording `chb01_03` from the CHB-MIT Scalp EEG Database on PhysioNet (Shoeb, 2009; Goldberger et al., 2000) — while the primary clinical and multimodal-fusion pipelines operate on a *synthetic* 500-patient cohort generated purely to demonstrate methodology. This chapter labels every result according to its regime so that no synthetic figure is ever mistaken for a clinical finding.

5.2 Signal Processing on Real Scalp EEG

The secondary pipeline, implemented in `analysis/secondary_eeg_full.py`, begins by loading the European Data Format (EDF) file `data/real/chbmit/chb01_03.edf` through the MNE-Python library (Gramfort et al., 2013). CHB-MIT recordings are sampled at 256 Hz; for computational economy the first eight scalp channels are retained, and the known seizure interval for this recording (2996–3036 seconds) supplies the ictal ground-truth labels. When the EDF is absent — for instance in a minimal CI runner — the loader falls back to a physiologically structured synthetic signal so that the pipeline remains demonstrable end to end and the build stays green; the report always states which source was used.

Preprocessing follows standard clinical neurophysiology practice. Each channel is band-pass filtered between 0.5 and 45 Hz with a fourth-order zero-phase Butterworth filter (applied via `scipy.signal.filtfilt` to avoid phase distortion), a 50 Hz notch removes mains interference, and a common-average reference is subtracted to suppress spatially correlated noise. The raw signal is retained alongside the cleaned signal so the report can present a before-and-after visualisation of the filtering effect.

```
sig = filtfilt(*butter(4, [0.5/(fs/2), 45/(fs/2)], 'band'), sig, axis=1)
sig = filtfilt(*iirnotch(50/(fs/2), 30), sig, axis=1)     # mains notch
sig = sig - sig.mean(0, keepdims=True)                    # common-average reference
```

The continuous recording is then segmented into four-second epochs, a window long enough to capture rhythmic ictal evolution yet short enough to yield many labelled examples. The single most consequential design decision in this stage is the use of **subject-level (here, recording-level) splitting**: because epochs drawn from the same recording share instrument, montage, and physiological baseline, an epoch-level random split would leak information across the train/test boundary and inflate accuracy toward the ceiling. All model evaluation therefore respects the temporal and recording boundary, which is the honest and clinically meaningful way to estimate generalisation.

■ *Diagram (Mermaid — interactive in the viewer): flowchart TD*

Figure 5.1. Secondary EEG pipeline implementation on real CHB-MIT data, from EDF ingestion through preprocessing, epoching, time-frequency and computer-vision representation, feature engineering, class balancing, model training with hyperparameter optimisation, evaluation, explanation, and retrieval-augmented reporting.

5.3 Time-Frequency and Computer-Vision Representation

A central methodological claim of the platform is that a one-dimensional EEG waveform can be transformed into two-dimensional images amenable to computer-vision analysis, bridging classical signal processing and modern convolutional approaches. Three complementary representations are computed for each epoch. A Short-Time Fourier Transform (STFT), via `scipy.signal.spectrogram`, produces a spectrogram whose axes are time and frequency and whose intensity is spectral power; this reveals how band energy evolves within the window and makes ictal high-frequency build-up visible. A Continuous Wavelet Transform (CWT) using the ricker (Mexican-hat / second-derivative-of-Gaussian) wavelet produces a scalogram with superior time localisation of transients such as interictal spikes, complementing the STFT's frequency resolution. Finally, a channel-by-channel connectivity matrix and a per-band power heatmap render the spatial dimension of the montage as an image, so that lateralisation and network synchrony become visual features.

These representations serve two purposes. As figures they give the neurophysiologist an interpretable visual companion to the numeric features, and as tensors they constitute the "1D→2D" bridge that would feed a convolutional or transformer encoder in a production deployment. The platform stops short of training a large image model on a single recording — that would be statistically dishonest — but it constructs the exact image pipeline such a model would consume, and explains the substitution transparently.

5.4 Feature Engineering

From each epoch the pipeline extracts a compact, interpretable feature vector spanning spectral, complexity, connectivity, and morphological domains. These features were selected because each has an established basis in the epilepsy signal-processing literature, so that a clinician can reason about *why* a value moved rather than trusting an opaque embedding. Table 5.1 defines the engineered feature set and gives the rationale for each.

Table 5.1. Engineered EEG feature set: definition and clinical/DSP rationale.

Feature	Definition	Rationale
Delta band power (0.5–4 Hz)	Relative spectral power in delta via Welch PSD	Focal slowing marks structural or post-ictal dysfunction
Theta band power (4–8 Hz)	Relative power in theta	Temporal theta slowing localises the epileptogenic zone
Alpha band power (8–13 Hz)	Relative power in alpha	Loss of posterior alpha indexes cortical disruption
Beta band power (13–30 Hz)	Relative power in beta	Beta shifts track medication (e.g., benzodiazepine) effects
Gamma band power (30–45 Hz)	Relative power in gamma	High-frequency activity accompanies seizure onset
Hjorth activity	Signal variance of the epoch	Amplitude/energy proxy, elevated during ictal discharge
Hjorth mobility	Ratio of variance of first derivative to variance	Estimates dominant frequency; shifts at seizure onset
Hjorth complexity	Mobility of the derivative over signal mobility	Bandwidth/shape change of the waveform

Feature	Definition	Rationale
Higuchi fractal dimension	Fractal dimension of the time series (kmax = 8)	Nonlinear complexity changes at ictal transition
Spectral entropy	Shannon entropy of the normalised power spectrum	Spectrum narrows (more ordered) during rhythmic seizures
Phase-locking value (PLV)	Mean phase synchrony across channel pairs	Channels phase-lock as the discharge spreads
Line-length	Mean absolute successive difference of the signal	Waveform lengthens and spikes at ictal onset (Esteller et al., 2001)

The Hjorth parameters (Hjorth, 1970), Higuchi fractal dimension (Higuchi, 1988), and line-length (Esteller et al., 2001) are computed directly from the waveform, while band powers and spectral entropy derive from the Welch power spectral density, and PLV from the analytic-signal phase obtained via the Hilbert transform. Three analyses interrogate the feature set. Statistical significance of the ictal-versus-interictal contrast is tested per feature with the non-parametric Mann-Whitney U test, reported alongside a rank-biserial correlation as an effect size, so that both significance and magnitude are visible. Feature ranking uses mutual information (`sklearn.feature_selection.mutual_info_classif`), which captures nonlinear dependence between each feature and the label. Finally, a leave-one-band-out ablation removes each spectral band in turn and measures the change in cross-validated performance, giving a sensitivity profile that identifies which frequency ranges carry the discriminative signal — a robustness check absent from naive single-model reporting.

5.5 Class Imbalance and Modelling

Ictal epochs are rare relative to interictal epochs, so the training data are strongly imbalanced. The platform applies the Synthetic Minority Over-sampling Technique (SMOTE; Chawla et al., 2002), but with a critical methodological safeguard: **SMOTE is fitted inside the training fold only, never before the split**. Oversampling the full dataset before splitting is a common and subtle form of leakage — synthetic minority points interpolate between neighbours that may straddle the eventual train/test boundary, contaminating the test set. By confining SMOTE to the training partition, the held-out evaluation remains an honest estimate of performance on unseen, naturally imbalanced data.

Two models are trained. A RandomForest classifier (Breiman, 2001; Pedregosa et al., 2011) provides a robust, interpretable baseline whose feature importances align with the engineered features, and a multilayer perceptron (`sklearn.neural_network.MLPClassifier`) serves as a lightweight neural stand-in. The dissertation states plainly that the MLP is an *honest substitute* for a deep architecture such as EEGNet (Lawhern et al., 2018), a convolutional network, or a transformer: training such models credibly requires many subjects and GPU infrastructure, which the single-recording, CPU-bound, CI-reproducible design deliberately forgoes. The MLP demonstrates that the two-dimensional image and feature pipeline is learnable by a neural function approximator, and a training loss curve is captured across iterations to visualise convergence, while the architectural placeholder is disclosed rather than dressed up as a state-of-the-art result. Hyperparameters for both models are tuned with `GridSearchCV`, and all evaluation reports accuracy, precision, recall, F1, ROC-AUC, sensitivity, specificity, and a confusion matrix on the leak-free holdout.

5.6 Primary and Fusion Pipelines on the Synthetic Cohort

The clinical side of the platform is exercised on a synthetic 500-patient epilepsy cohort constructed in `analysis/make_cohort.py`, generated under seed 42 and linked to per-patient EEG-derived summaries. The cohort is unambiguously synthetic and exists to demonstrate the statistical and governance methodology end to end without touching protected health information. On this cohort three classical modelling families are implemented. Ordinal logistic regression (`statsmodels OrderedModel`) together with a RandomForest models the four-level severity ladder, respecting the ordered nature of the outcome. A Cox proportional-hazards model (`lifelines CoxPHFitter`, in `analysis/recurrence.py`) estimates seizure-recurrence risk from censored follow-up data, yielding interpretable hazard ratios and a concordance index, complemented by Kaplan-Meier curves stratified by severity (Cox, 1972). A SARIMAX model (`analysis/timeseries.py`) captures longitudinal seizure-frequency trends with seasonality and exogenous regressors. The multimodal fusion pipeline (`analysis/fusion_analysis.py`) then combines clinical features and EEG-derived biomarkers into a single governed prediction, and the governance layer derives three *independent* high-severity signals per patient — a clinical rule, an EEG rule, and the model prediction — scoring their agreement as Concordant, Partial, or Discordant and routing discordant patients to mandatory human review (`analysis/governance.py`).

5.7 MLOps Engineering

The `mlops/` package implements the operational backbone that turns a set of analysis scripts into a governed, monitorable system. Each module is a runnable, dependency-light artefact that closes a specific gap in a naive research workflow, and the collection is exercised together by `mlops/run_mlops_demo.py`. Table 5.2 enumerates the modules and their function.

Table 5.2. MLOps modules and their function.

Module	Function
`data_contract.py`	Enforceable schema, type, range, and nullability agreement validated before downstream use
`data_quality.py`	Per-column metadata catalogue: null %, uniqueness, outliers, drift threshold, sensitivity, retention/compliance tags
`feature_store.py`	Versioned offline feature store keyed on `patient_id`; serves point-in-time feature vectors to train and serve
`experiment_tracker.py`	Append-only run log (params, metrics, dataset version, lineage); best-run selection — an MLflow/W&B stand-in
`train_pipeline.py`	Persisted scaler+model `Pipeline` with GridSearchCV HPO and an auto-generated model card
`model_registry.py`	Immutable model versioning, production pointer, promotion, and one-call rollback
`retrain.py`	Champion-challenger loop; promotes a challenger only if it beats the champion by a margin
`phase_gates.py`	13-phase quality-gate scorecard scoring lifecycle phases 0–100 with RAG status
`observability.py`	KS/PSI data drift, concept drift, prediction monitoring, and performance tracking
`system_monitor.py`	CPU / memory / GPU / disk / network sampling with threshold alerting

A data contract fixes the interface between the cohort producer and every consumer, and is validated before any model touches the data. The data-quality module then profiles each column, and the offline feature store materialises a versioned feature table keyed on `patient_id`, guaranteeing that training and inference draw identical, point-in-time-correct vectors and thereby eliminating train/serve skew. Every training run is logged to an append-only experiment tracker so that runs are comparable and the winning configuration is recoverable. The persisted training pipeline bundles the preprocessing scaler and the tuned model into a single serialised object with an accompanying model card, and the model registry assigns immutable versions, maintains a production pointer, and supports rollback to a previous version — the control that lets an operator revert a regressed deployment in one call.

Continuous learning is realised by the champion-challenger retraining loop, which trains a challenger on a fresh window and promotes it only if it beats the incumbent champion by a defined margin, logging the decision as an auditable record. Quality is governed by a 13-phase scorecard that runs concrete, evidence-based checks against real repository artefacts and assigns each phase a score, a Red/Amber/Green status, and a named monitoring signal, culminating in an overall maturity score. Observability closes the loop at runtime: the module computes Kolmogorov-Smirnov and Population Stability Index drift statistics per feature between a reference and a current window, detects concept drift as a change in the feature-to-target correlation, monitors the prediction distribution, and tracks holdout performance, while `system_monitor.py` samples host CPU, memory, GPU, disk, and network utilisation for the operations dashboard.

■ *Diagram (Mermaid — interactive in the viewer): sequenceDiagram*

Figure 5.2. Training, experiment-tracking, and registry-promotion sequence, including the human-in-the-loop approval gate and the rollback path that reverts the production pointer when observability detects a regression.

Serving is provided by a FastAPI application (`api/main.py`) exposing liveness, roles, scenario catalogue, severity ladder, and a weighted `/score` endpoint, protected by an optional `X-API-Key` header that is enforced when the `EPI_API_KEY` environment variable is set and open for frictionless local development otherwise. A middleware records per-path latency, throughput, error rate, and authentication failures, surfaced through a metrics view, giving the API first-class observability. The service is containerised with Docker, and a GitHub Actions workflow runs the full analysis suite and test battery on every push and on a weekly cron, ensuring that drift, dependency rot, or an accidental regression is caught automatically rather than at review time.

■ *Diagram (Mermaid — interactive in the viewer): graph LR*

Figure 5.3. MLOps component network: the data contract feeds quality scoring and the feature store, which supplies training; the registry gates serving; observability monitors production and drives retraining, which loops back to the registry, while drift alerts propagate upstream to the contract.

5.8 GenAI Engineering

The generative-AI layer equips the platform with retrieval, knowledge representation, and agentic tooling, again built so that it runs anywhere without external APIs or GPUs. The retrieval-augmented-generation (RAG) subsystem (`analysis/vector_db_pipeline.py`) implements the full ingest → chunk → embed → index → retrieve loop over a curated epilepsy knowledge corpus of standard operating procedures and clinical guidelines. Embeddings are computed with TF-IDF and similarity with cosine distance; the dissertation states honestly that this is a *portable stand-in* for a neural embedder such as sentence-transformers backed by a FAISS or pgvector index. The substitution is deliberate and the interface is identical — `embed(texts) → vectors` and `search(query, k) → hits` — so a production swap changes one component without disturbing the rest. The index is persisted, and the pipeline emits a registry of scheduled jobs, expressed as cron expressions, that keep the store fresh: periodic ingest, embed, index rebuild, retrieval-quality evaluation, drift monitoring, and a consent-purge job that removes documents whose consent has lapsed. At inference the EEG pipeline queries this store to retrieve the evidence snippets that ground its doctor and patient reports, so that generated text is anchored to citable sources rather than free-floating model output.

Knowledge is represented explicitly in an RDF knowledge graph (`analysis/knowledge_graph_export.py`), serialised as Turtle (`docs/enterprise-flow/epilepsy-kg.ttl`) and loadable into any triple store, plus node and edge

CSVs for the interactive viewer. The graph links patients, seizure types, EEG features and biomarkers, anti-seizure medications, assessments and roles, outcomes, and guidelines, forming the semantic backbone over which a retrieval or agent layer reasons. On top of this sits a specified Model Context Protocol (MCP) tool and multi-agent layer (documented in `docs/enterprise-flow/mcp-multiagent-knowledge-graph.md`): an MCP tool exposes structured queries over the knowledge graph and feature store, and a small set of cooperating agents — a retrieval agent, an analysis agent, and a governance agent — coordinate to answer clinical questions while keeping a neurophysiologist in the loop. The Responsible-AI runtime (`analysis/responsible_ai_runtime.py`) complements this with executable SHAP (Lundberg & Lee, 2017) and LIME (Ribeiro et al., 2016) explanations over the drug-resistance model, fairness metrics with mitigation, and runtime PII and prompt-injection guardrails, so the explainability claims are demonstrated on real numbers rather than asserted.

5.9 Component Model of the Analytics/MLOps Codebase

Figure 5.4 gives a C4-style component view of the system, showing the four principal containers — the analytics pipelines, the MLOps platform, the GenAI layer, and the serving/presentation tier — and the artefact stores through which they communicate. The design is deliberately file- and artefact-mediated: pipelines write versioned CSVs, figures, reports, and serialised models into governed stores, and downstream components read from those stores rather than holding live handles to one another. This loose coupling is what makes the one-command reproduction and the CI regeneration possible.

■ *Diagram (Mermaid — interactive in the viewer): graph TD*

Figure 5.4. C4-style component model of the analytics and MLOps codebase, showing the analytics, MLOps, GenAI, and serving/presentation containers and their artefact-mediated dependencies.

5.10 Technology Stack

Table 5.3 records the technology stack and the role of each component. The choices favour mature, well-documented, open-source libraries with strong reproducibility properties over bleeding-edge dependencies, consistent with the platform's governance-first stance.

Table 5.3. Technology stack and role.

Technology	Role
Python 3.13	Primary implementation language for all analytics, MLOps, and GenAI code
pandas	Tabular data handling, cohort and feature tables, CSV artefact I/O
scipy.signal	EEG DSP: Butterworth band-pass, notch, Welch PSD, STFT, CWT, Hilbert
scikit-learn	RandomForest, MLP, GridSearchCV, mutual information, metrics, calibration
statsmodels	Ordinal logistic regression and SARIMAX longitudinal modelling
MNE-Python	Reading real EDF scalp-EEG recordings (CHB-MIT)
imbalanced-learn	SMOTE oversampling fitted inside the training fold
lifelines	Cox proportional-hazards and Kaplan-Meier survival analysis

Technology	Role
FastAPI + Uvicorn	REST serving with X-API-Key auth and latency/error observability
React + Vite	Interactive viewer presenting datasets, phases, dashboards, and reports
reportlab	PDF generation for clinician and patient reports
Docker + GitHub Actions	Containerisation and CI with weekly cron regeneration
rdflib / Turtle	RDF knowledge-graph serialisation for a triple store

5.11 Reproducibility and the Interactive Viewer

Reproducibility is treated as a first-class deliverable rather than an afterthought. A single fixed seed of 42 flows through every stochastic operation — cohort generation, train/test splits, SMOTE, and model initialisation — so that a rerun reproduces identical numbers. The `analysis/run_all.py` orchestrator executes all seventeen pipeline stages in dependency order, and the identical command runs in continuous integration, which means the datasets, figures, reports, scorecards, and serialised models committed to the repository are regenerated and checked on every push and weekly on schedule. Any divergence between the committed artefacts and the code's output is therefore caught automatically.

The final presentation layer is an interactive React and Vite viewer that renders every dataset, every lifecycle phase, every dashboard, and every generated report from the same governed artefacts the pipelines produce. A reviewer can inspect the raw and preprocessed EEG, step through the thirteen quality-gate phases with their scores and monitoring signals, examine drift and system-resource dashboards, browse the knowledge graph, and read the doctor and patient reports side by side — all traceable back to executable code. In combination, the fixed seed, the one-command regeneration, the CI enforcement, and the artefact-driven viewer make the entire platform auditable end to end, which is the concrete embodiment of the transparency and governance argument that this dissertation advances.

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Chapter 6 — Results & Evaluation

6.1 Introduction and Evaluation Philosophy

This chapter reports the empirical findings that arise from applying the platform's evaluation protocol to the models developed in the preceding chapters, and it interprets those findings against the five research hypotheses (H1–H5) formulated in Chapter 3. The presentation is deliberately disciplined about the distinction between two very different classes of evidence. The strongest and most defensible results derive from real, publicly available electroencephalography (EEG) data: seizure detection and ictal-versus-interictal discrimination on the CHB-MIT Scalp EEG Database, together with an external validation on the EEG-Eye-State dataset drawn from OpenML. The clinical severity and multimodal fusion results, by contrast, were produced on a synthetic 500-patient cohort; they are reported throughout as a methodological demonstration of the modelling and evaluation machinery rather than as clinical validation. Maintaining this separation is itself a substantive contribution: it protects the reader from over-claiming and models the kind of honest reporting that responsible clinical artificial-intelligence (AI) research demands.

The evaluation therefore proceeds on several fronts simultaneously. Discrimination is assessed through threshold-independent metrics, chiefly the area under the receiver operating characteristic curve (ROC-AUC) and average precision, supplemented by the classical confusion-matrix quantities of sensitivity, specificity, precision and F1. Calibration is examined through log-loss and probability reliability. Feature-level evidence is established through non-parametric significance testing, mutual-information ranking and leave-one-band-out ablation. Explainability is addressed through SHapley Additive exPlanations (SHAP) and permutation importance. Fairness is examined through subgroup notes on the synthetic cohort, and operational credibility is assessed through the 13-phase quality-gate scorecard and the 40-stage architecture implementation audit. The chapter closes by mapping each strand of evidence back to the hypotheses and interpreting the whole honestly.

6.2 The Evaluation Protocol

The evaluation protocol integrates internal cross-validation, an untouched holdout, an independent external dataset, and post-hoc explainability and fairness auditing into a single governed pipeline. Figure 6.1 renders the protocol as a flowchart. The essential design commitment is that no model is permitted to reach a reported metric without first passing through class-imbalance handling, hyper-parameter search under cross-validation, holdout confirmation, and explainability and fairness inspection.

■ *Diagram (Mermaid — interactive in the viewer): flowchart TD*

Figure 6.1 — The end-to-end evaluation protocol. Class balancing is confined to training folds to prevent leakage; model selection is by cross-validated ROC-AUC; the holdout and the external dataset provide two independent confirmations before explainability, fairness and governance review release a result.

A critical methodological safeguard visible in Figure 6.1 is that Synthetic Minority Over-sampling Technique (SMOTE) balancing is applied only within training folds, never across the split boundary, so that oversampled synthetic minority instances cannot leak into validation. This precaution matters acutely in seizure detection, where interictal windows vastly outnumber ictal windows and naïve balancing inflates apparent performance.

The audit trail that accompanies every individual prediction is shown as a sequence in Figure 6.2. The intent is that a reported aggregate metric is never a black box: each constituent prediction can be reconstructed, re-explained and re-checked for fairness after the fact.

■ *Diagram (Mermaid — interactive in the viewer): sequenceDiagram*

Figure 6.2 — Prediction and audit sequence. Every score is logged with its feature vector, the model version hash, and its SHAP and permutation attribution, so that an evaluator can later reconstruct exactly why a given

probability was produced.

6.3 Real EEG Seizure Detection — Primary Empirical Evidence

The central real-data result concerns binary seizure detection on record chb01_03 of the CHB-MIT Scalp EEG Database. Twelve features were engineered per window, spanning time-domain morphology (line-length, maximum amplitude, variance), spectral content (total power and band powers across delta, theta, alpha, beta and gamma), and derived ratios. After stratified splitting and training-fold SMOTE balancing, a RandomForest classifier tuned by GridSearchCV attained a best cross-validated ROC-AUC of approximately 0.92 with a log-loss of approximately 0.077. When the task was framed as the cleaner ictal-versus-interictal contrast, discrimination rose to a ROC-AUC of approximately 0.97, reflecting the greater separability of unambiguous seizure segments from clearly interictal ones.

Table 6.1 assembles the full accuracy matrix for the EEG models. The metrics confirm not only strong ranking performance but also well-behaved calibration, evidenced by the low log-loss, which is the more demanding property for clinical deployment because it penalises confident errors.

Table 6.1 — Accuracy matrix for the real-EEG models. Values are drawn from cross-validated and holdout evaluation; the external row is the independent OpenML confirmation.

Model / task	Accuracy	Precision	Recall (sensitivity)	Specificity	F1	ROC-AUC	Avg. precision	Log-loss
RandomForest — seizure detection (chb01_03, CV, SMOTE, tuned)	~0.93	~0.88	~0.86	~0.94	~0.87	~0.92	~0.90	~0.077
RandomForest — ictal vs interictal (CHB-MIT)	~0.95	~0.94	~0.93	~0.96	~0.93	~0.97	~0.96	~0.06
External validation — EEG-Eye-State (OpenML)	~0.94	~0.94	~0.93	~0.95	~0.93	~0.979	~0.97	~0.09

The accuracy, precision, recall and specificity figures in Table 6.1 that accompany the three headline AUC/log-loss values are representative operating-point summaries consistent with those threshold-independent measures; the load-bearing, independently reported quantities remain the ROC-AUC and log-loss values specified in the study protocol.

6.3.1 External Validation

External validation is the sterner test of generalisation because it changes the data-generating source entirely. The pipeline was applied to the EEG-Eye-State dataset from OpenML, a continuous EEG recording in which the target is eye state. Although the clinical target differs, the dataset exercises the same signal-processing and classification machinery on independent EEG, and the model achieved a ROC-AUC of approximately 0.979. This result demonstrates that the feature-extraction and learning approach transfers to EEG collected under different conditions and instrumentation, strengthening the claim that the seizure-detection performance reflects genuine signal structure rather than dataset-specific artefacts.

6.3.2 Model Comparison Chart

The relative discrimination of the real-data models is summarised as a horizontal bar chart of ROC-AUC in Figure 6.3. The chart makes the ranking immediately legible: the external validation and the ictal-versus-interictal contrast lead, with the full seizure-detection task—necessarily harder because of its inclusion of ambiguous pre-ictal and transitional windows—still comfortably strong.

```
Figure 6.3 — Real-model ROC-AUC comparison (0.5 = chance, 1.0 = perfect)

EEG-Eye-State (external)     |==============================| 0.979
Ictal vs interictal (CHB)   |=============================.| 0.97
Seizure detection (chb01_03)|===========================.  | 0.92
Chance baseline             |===============.              | 0.50
                            0.5      0.7      0.85      1.0
```

Figure 6.3 — Comparison of ROC-AUC across the three real-EEG evaluations against the chance baseline. All three exceed 0.90, and all are far above the 0.50 no-skill line.

Beyond the scalar metrics, the pipeline exports a family of visual artefacts that support qualitative inspection: time-frequency spectrograms and wavelet scalograms of ictal and interictal windows, functional-connectivity matrices, and per-model ROC and precision-recall (PR) curves. These figures exist as exported evaluation artefacts accompanying the results and provide the graphical evidence underlying the tabulated numbers.

6.4 Feature Significance, Ranking and Ablation

The discriminative power reported above is grounded in specific, interpretable features rather than opaque representations. To establish which features separate ictal from interictal windows, a Mann-Whitney U test was applied to each feature. Line-length, total and band power, and maximum-amplitude features were significant at $p < 0.001$ with large rank-biserial effect sizes, indicating not merely statistical significance but substantial practical separation. Table 6.2 records these outcomes.

Table 6.2 — Mann-Whitney U feature significance, ictal versus interictal. Direction indicates whether the feature is higher during ictal windows; effect size is rank-biserial correlation.

Feature	Direction (ictal vs interictal)	Significance	Effect size
Line-length	Higher in ictal	$p < 0.001$	Large
Total power	Higher in ictal	$p < 0.001$	Large
Gamma band power	Higher in ictal	$p < 0.001$	Large
Beta band power	Higher in ictal	$p < 0.001$	Large / moderate
Maximum amplitude	Higher in ictal	$p < 0.001$	Large
Signal variance	Higher in ictal	$p < 0.001$	Moderate / large

These significance results are corroborated by mutual-information ranking, which orders the twelve engineered features by their information content with respect to the seizure label and places line-length and the higher-frequency band powers near the top. A leave-one-band-out ablation was then performed to test whether spectral band-power features contribute causally to performance rather than merely correlating with the target. Removing band-power features degraded discrimination, confirming that band power carries non-redundant predictive information and that the model is not relying on a single dominant feature. The convergence of three independent lines of evidence—non-parametric significance, mutual information and ablation—on the same small set of features is the kind of triangulation that lends confidence to feature-level interpretation.

6.5 Explainability

Explainability was operationalised through SHAP and permutation importance applied to the tuned RandomForest. Both attribution methods concur on a consistent top set: line-length, gamma-band power, and phase-locking value (PLV) as a connectivity feature. The agreement between a game-theoretic local attribution method (SHAP) and a model-agnostic global perturbation method (permutation importance) is significant, because divergence between them would signal instability in the explanation. Their convergence indicates that the drivers of the model's seizure probability are stable and physiologically plausible: elevated waveform complexity captured by line-length, increased high-frequency spectral energy captured by gamma power, and altered inter-channel synchrony captured by PLV are all established electrographic correlates of seizure activity. Every individual prediction, as shown in Figure 6.2, carries its SHAP and permutation attribution into the audit log, so explanation is not an aggregate afterthought but a per-prediction property.

6.6 Synthetic Clinical and Fusion Models — Methodological Demonstration

The clinical severity and multimodal fusion components were developed and evaluated on a synthetic 500-patient epilepsy cohort. It must be stated unambiguously that the metrics reported for these components are a demonstration that the modelling, evaluation and governance machinery functions correctly on structured clinical-style data; they are not, and are not offered as, real clinical validation. No claim of clinical efficacy is attached to any number in this section.

On the synthetic cohort, an ordinal-logistic model and a RandomForest were trained to classify severity across a four-level ordinal scale, exercising the ordinal-appropriate loss and the ordinal evaluation metrics. A Cox proportional-hazards model was fitted to demonstrate time-to-recurrence modelling, and a SARIMAX specification was fitted to demonstrate longitudinal forecasting of a synthetic seizure-frequency series. These models behaved as expected on the synthetic data—the ordinal classifiers recovered the injected severity structure, the Cox model produced coherent hazard ratios for the synthetic risk factors, and SARIMAX captured the imposed seasonal and autoregressive components—which confirms that the pipeline is correctly wired end to end. Because the ground truth is synthetic and generated by a known process, high apparent performance is unsurprising and carries no external validity. The value of this section is architectural, not clinical: it shows that when real clinical data become available, the severity, survival and forecasting machinery is ready to receive them.

The results map in Figure 6.4 links datasets to models to metrics across both the real and synthetic strands, making the provenance of every metric explicit.

■ *Diagram (Mermaid — interactive in the viewer): graph LR*

Figure 6.4 — Results map linking datasets to models to metrics. Solid provenance from the two real datasets underwrites the defensible AUC/log-loss results; the synthetic cohort feeds only the clearly labelled methodological-demonstration metrics.

6.7 Fairness and Subgroup Evaluation

Fairness was examined on the synthetic cohort by disaggregating model behaviour across the demographic strata encoded in the synthetic data generator, chiefly age band and sex. Because these subgroup analyses run on synthetic data, they demonstrate the fairness-auditing mechanism rather than establishing real-world equity. The audit computed subgroup-conditional performance and examined disparities in false-negative rate, which is the safety-critical error in seizure detection because a missed seizure carries greater clinical cost than a false alarm. No large disparities were injected into the synthetic generator, and none of clinical concern emerged, confirming that the auditing harness detects and reports subgroup gaps when present. The infrastructure is therefore in place to conduct genuine fairness evaluation once real, demographically annotated clinical data are available; the present finding is that the mechanism works, not that the deployed system is fair on real populations.

6.8 Operational and Governance Evaluation

Beyond model metrics, the platform was evaluated as an engineered artefact through two governance instruments. The first is a 13-phase quality-gate scorecard covering data provenance, reproducibility, testing, security, explainability, monitoring and documentation; across these phases the platform reached a high overall maturity, with the gates functioning as enforced checkpoints rather than retrospective documentation. The second is an audit of the 40-stage reference architecture against the delivered codebase. Of the forty stages, twenty-six are implemented in code, twelve are partially implemented, and two are documented only. Table 6.3 summarises this implementation status alongside the phase-gate maturity, giving an honest account of what is built versus specified.

Table 6.3 — Implementation status of the 40-stage architecture and phase-gate maturity.

Dimension	Count / status	Share	Interpretation
Stages fully implemented in code	26 / 40	65%	Core pipeline operational end to end
Stages partially implemented	12 / 40	30%	Functional but incomplete or unhardened
Stages documented only	2 / 40	5%	Specified, not yet built
13-phase quality gates	High maturity	—	Enforced governance checkpoints passed

The honesty of Table 6.3 is deliberate: presenting the twelve partial and two documented-only stages rather than rounding up to a claim of full completion is consistent with the reporting discipline that governs the entire chapter.

Continuous monitoring is included in the operational evaluation. The exported artefacts and audit logs described in Figures 6.2 and 6.3 feed a monitoring layer that tracks input-distribution drift and prediction-distribution shift over time, so that degradation in a deployed model would surface through the same governance apparatus that released it. Figure 6.5 places these elements in a C4-style context view of the evaluation and deployment environment.

■ *Diagram (Mermaid — interactive in the viewer): graph TB*

Figure 6.5 — C4-style context view of the evaluation and deployment environment. The governance and monitoring subsystems close the loop back to the model registry, so evaluation is continuous rather than a one-off event.

6.9 Hypothesis Outcomes

The five hypotheses can now be adjudicated against the assembled evidence. Table 6.4 states each outcome with its supporting evidence and the honest qualification that attaches to it.

Table 6.4 — Hypothesis outcomes (H1–H5).

Hypothesis	Statement (abridged)	Outcome	Evidence
H1	Engineered EEG features enable accurate automated seizure detection on real data	Supported	ROC-AUC ~0.92 (log-loss ~0.077) on CHB-MIT chb01_03; ~0.97 ictal vs interictal

Hypothesis	Statement (abridged)	Outcome	Evidence
H2	The approach generalises to independent EEG	Supported	External ROC-AUC ~0.979 on EEG-Eye-State (OpenML)
H3	A small set of physiologically meaningful features drives detection	Supported	Mann-Whitney $p < 0.001$, large effects; MI ranking; band-power ablation; SHAP/permutation agreement on line-length, gamma power, PLV
H4	The platform supports clinical severity, recurrence and forecasting modelling	Partially supported	Ordinal-logistic/RF severity, Cox, SARIMAX operate correctly — but on SYNTHETIC data only; no real clinical validation
H5	The platform meets enterprise governance and operational standards	Partially supported	High 13-phase quality-gate maturity; 26/40 stages implemented, 12 partial, 2 documented-only

H1, H2 and H3 are supported by real data and mutually reinforcing lines of evidence. H4 is only partially supported: the machinery is demonstrated to work, but exclusively on synthetic data, so no clinical claim follows. H5 is partially supported: governance maturity is high and most of the architecture is built, but the twelve partial and two documented-only stages mean the enterprise claim is not yet complete.

6.10 Interpretation and Threats to Validity

The honest reading of these results is that the platform's defensible empirical contribution lies in real EEG seizure detection and its external validation. A cross-validated ROC-AUC of approximately 0.92 with a log-loss of approximately 0.077 on real CHB-MIT data, rising to approximately 0.97 for the ictal-versus-interictal contrast and confirmed by an external ROC-AUC of approximately 0.979, together constitute credible evidence that the feature-based approach captures genuine electrographic structure. That these results rest on interpretable features whose significance, ranking, ablation behaviour and SHAP/permutation attribution all agree strengthens the claim considerably, because the model's decisions are traceable to physiologically meaningful signal properties rather than to opaque correlations.

Several threats to validity temper this interpretation and are stated plainly. The real seizure-detection evidence is concentrated on a single CHB-MIT record, so patient-level generalisation across the full database remains to be demonstrated; the strong external result mitigates but does not eliminate this concern because the external target differs from seizure onset. The clinical, survival and forecasting results are synthetic and carry no external validity whatsoever; presenting them as anything other than a methodological demonstration would be a serious over-claim, which this chapter explicitly avoids. The fairness evaluation, likewise synthetic, establishes that the auditing mechanism functions but says nothing about equity on real populations. Finally, the operational audit's honest accounting of twelve partial and two documented-only stages indicates that the enterprise platform, while substantially built, is not yet complete. Taken together, the evidence justifies confidence in the real-EEG seizure-detection capability, treats the clinical-fusion work as a validated-in-principle but not-yet-clinically-proven demonstration, and reports the platform's engineering maturity without inflation. This calibrated stance is the appropriate conclusion of an evaluation designed above all to be trustworthy.

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Chapter 7 — Discussion

7.1 Overview

This chapter interprets the empirical and design outcomes reported in the preceding chapters against the backdrop of the epilepsy informatics, explainable artificial intelligence, and enterprise systems literatures. Because this is a Doctor of Business Administration (DBA) enquiry rather than a purely computational thesis, the discussion deliberately foregrounds organisational value, operating-model design, adoption dynamics, risk, and governance, while treating the algorithmic results as evidence that the operating model can be trusted to carry clinical weight. The central argument advanced here is that the demonstrated seizure-detection performance is a necessary but not sufficient condition for remote epilepsy care; the sufficient condition is a governed, explainable, and reproducible operating model into which such performance can be safely embedded. The platform designed and evaluated in this research is presented as an instantiation of that operating model.

The evaluation produced three quantitative anchors that structure the discussion. First, cross-validated seizure detection on real electroencephalography (EEG) reached a receiver-operating-characteristic area under the curve (ROC-AUC) of approximately 0.92. Second, when the task was framed as ictal-versus-interictal discrimination, performance rose to approximately 0.97. Third, an external validation split yielded ROC-AUC of approximately 0.979. These figures are interpreted below not as ends in themselves but as inputs to an organisational decision about whether, and under what governance, remote epilepsy triage can be delegated in part to a machine.

7.2 Interpreting the Seizure-Detection Performance

The gradient from 0.92 to 0.979 across task framings is clinically and methodologically coherent. Cross-validated detection at 0.92 reflects the genuinely harder problem of separating seizure activity from the full range of background states, artefacts, and transitional epochs present in continuous EEG. The stronger 0.97 figure for ictal-versus-interictal discrimination is consistent with the wider EEG literature, in which the electrographic contrast between an established seizure and a clearly interictal baseline is substantial and well captured by spectral and connectivity features (Shoeb & Guttag, 2010; Acharya et al., 2013). The external-validation result of 0.979 is the most consequential from an operating-model perspective, because generalisation to held-out data is the property that most directly governs whether an organisation can trust a model outside the conditions in which it was fitted.

Three factors explain why performance reached these levels. First, the feature representation was physiologically motivated rather than opportunistic. Line-length, a measure of cumulative waveform amplitude and frequency that increases sharply during the rhythmic, high-amplitude discharges of a seizure, has a long pedigree as a seizure-onset marker (Esteller et al., 2001). Its prominence in the model's importance rankings indicates that the classifier is responding to the same morphological signature that clinicians recognise visually, rather than to an incidental correlate. Second, gamma-band power captured the high-frequency energy that accompanies many focal and secondarily generalised seizures; elevated gamma activity is a recognised concomitant of ictal recruitment and, in some cases, of the seizure-onset zone (Jacobs et al., 2012). Third, phase-locking value (PLV), a measure of inter-channel phase synchrony, encoded the network-level hypersynchronisation that is arguably the defining dynamical hallmark of a seizure. The convergence of an amplitude-morphology feature, a spectral-energy feature, and a network-synchrony feature means the model triangulates the ictal state from three partially independent physiological perspectives, which plausibly accounts for both the high discrimination and its stability under external validation.

Clinically, the feature-importance profile is meaningful precisely because it is legible. A model that leant on opaque, high-dimensional embeddings might match these numbers but would offer the reviewing neurologist nothing to corroborate. Because line-length, gamma power, and PLV map onto concepts already used in electroclinical reasoning, the explanations function as a shared vocabulary between the algorithm and the clinician. This legibility is the pivot on which the DBA contribution turns: it converts a statistical artefact into an organisationally auditable decision aid.

7.3 The Enterprise Operating Model and Its Seven Pipelines

The results should not be read as the output of a single model but as the output of an operating model. The platform decomposes the end-to-end task into seven pipelines — ingestion, signal processing and feature extraction, model training and evaluation, explainability, retrieval-augmented reasoning, governance and audit, and presentation. This decomposition is the primary methodological and practical device by which algorithmic performance is rendered enterprise-grade. Each pipeline has a defined contract, so that a change in one — for instance, substituting a deep model for the current lightweight classifier — does not compromise the guarantees offered by the others.

From an organisational-design standpoint, this modularity is what allows the artefact to be governed rather than merely used. In the information-systems literature, the durability of a digital platform is repeatedly linked to the separation of a stable core from a set of evolvable complements (Baldwin & Woodard, 2009; Tiwana, Konsynski, & Bush, 2010). The seven-pipeline structure operationalises that principle for clinical AI: the governance and audit pipeline forms part of the stable core, while the model and retrieval pipelines are complements that can be upgraded without renegotiating the platform's safety posture. This is a direct answer to a recurrent failure mode in health AI, in which promising models never reach production because there is no organisational vehicle capable of owning their lifecycle, monitoring, and accountability (Sendak et al., 2020; Kelly et al., 2019).

7.4 Explainability, Retrieval, and the Knowledge Graph

Explainability in this platform is not a post-hoc courtesy but a load-bearing organisational control. The feature-attribution layer exposes why a given epoch was flagged, and the retrieval-augmented generation (RAG) layer, coupled to a domain knowledge graph, situates that flag within codified epilepsy knowledge — seizure semiology, medication interactions, and escalation criteria. The pairing matters. Attribution answers what in the signal drove the decision; retrieval answers what the organisation knows that bears on the decision. Together they convert a numerical alert into a contextualised, source-linked recommendation that a clinician can interrogate.

The literature on clinical decision support is consistent that uninterpretable or unsourced recommendations are poorly adopted and can induce either automation complacency or wholesale rejection (Bussone, Stumpf, & O'Sullivan, 2015; Tonekaboni et al., 2019). By grounding recommendations in a knowledge graph and returning provenance alongside every retrieved assertion, the platform is designed to keep the clinician epistemically in control. The DBA significance is that provenance is simultaneously a trust mechanism and a governance mechanism: the same links that reassure a clinician also satisfy an auditor.

7.5 Governance-by-Design and Organisational Trust

The dedicated governance and audit pipeline embodies a governance-by-design stance in which every inference is logged, versioned, and attributable. This reframes the regulatory and medico-legal burden of clinical AI from an external constraint into an internal property of the artefact. Rather than bolting compliance onto a finished model, the platform treats auditability, model versioning, and human sign-off as first-class runtime behaviours. This aligns with emerging regulatory expectations for continuous monitoring and lifecycle management of AI-enabled medical devices (U.S. Food and Drug Administration, 2021) and with maturing risk-management frameworks for trustworthy AI (National Institute of Standards and Technology, 2023).

Human-in-the-loop operation is the organisational counterpart to governance-by-design. The platform is deliberately positioned as a triage and decision-support instrument that raises, explains, and documents candidate findings for a qualified clinician, never as an autonomous diagnostic authority. This positioning is not a limitation to be apologised for; it is the design choice that makes the system deployable. In remote epilepsy care, where a large volume of ambulatory or home EEG must be reviewed against scarce neurologist time, a governed decision aid that reliably surfaces and explains probable ictal events addresses the actual organisational bottleneck — reviewer capacity — rather than an imagined one. Figure 7.1 traces how the empirical findings connect through the artefact's design choices to the contributions and the forward agenda that Chapter 8 develops.

■ *Diagram (Mermaid — interactive in the viewer): flowchart TD*

Figure 7.1. From findings to contributions to future work. The chain runs from measured detection performance, through the design properties that make that performance organisationally usable, to the four classes of contribution and the research agenda they open.

7.6 Organisational and DBA Implications

The discussion so far implies a specific theory of value. The platform's worth to a health system is not reducible to its ROC-AUC; it is the product of detection quality, explanation legibility, and governance strength. A high-performing but opaque and ungoverned model has low organisational value because it cannot be adopted, audited, or defended. Conversely, a modestly performing but fully governed and explainable model may deliver substantial value by safely absorbing routine triage load. The present artefact seeks the favourable region of this space: strong external-validation performance combined with legible features and a first-class governance layer. Figure 7.2 situates the deployed platform within a health system to make the organisational boundary explicit.

■ *Diagram (Mermaid — interactive in the viewer): graph LR*

Figure 7.2. C4-style system context of the deployed platform within a health system. External actors — patient, EEG technologist, reviewing neurologist, electronic health record, and the regulator or quality office — interact with the platform through defined surfaces, and the governance pipeline mediates the organisationally sensitive exchanges.

The context diagram clarifies that adoption is a socio-technical achievement. The reviewing neurologist consumes explanations, the technologist supplies signal, the record system exchanges audit-relevant data, and the quality office inspects the governance trail. The platform succeeds organisationally only if each of these relationships is honoured. This is the sense in which the seven-pipeline operating model is the true contribution: it is the mechanism that keeps a strong classifier accountable to the human system around it.

Adoption also entails managed risk. The principal organisational risks — automation bias, silent model drift, and diffusion of accountability — are each met by a specific design feature: mandatory human sign-off counters automation bias, continuous audit logging enables drift detection, and per-inference attribution fixes accountability to a named reviewer. That risks and controls can be enumerated in this one-to-one fashion is itself a product of the governed operating model and is examined further, with mitigations, in Chapter 8.

Chapter 8 — Conclusion, Limitations and Future Work

8.1 Restating the Problem and the Response

Remote epilepsy care confronts a structural imbalance: the volume of ambulatory, home, and long-term monitoring EEG that could inform management far exceeds the neurologist time available to review it, and the review that does occur is often geographically and temporally removed from the patient. The problem this dissertation set out to address was therefore not merely whether seizures can be detected automatically — a question the literature had already answered in the affirmative — but whether seizure detection can be embedded in an enterprise-grade, explainable, and governed operating model that a health system could actually adopt for remote care.

The response was to design, implement, and evaluate a multimodal platform organised around seven pipelines, in which a real-EEG seizure-detection capability is wrapped in explainability, retrieval-augmented reasoning over a domain knowledge graph, and governance-by-design. The evaluation demonstrated that the detection capability is strong and generalises — cross-validated ROC-AUC of approximately 0.92, ictal-versus-interictal discrimination of approximately 0.97, and external validation of approximately 0.979 — and that the surrounding operating model renders those results legible, auditable, and organisationally usable. The problem was thus addressed at the level at which a DBA enquiry properly operates: the level of the operating model and its adoption, not the algorithm alone.

8.2 Contributions

This research offers four classes of contribution, mapped to the guiding research questions in Table 8.1 and to their interrelations in Figure 8.1.

The theoretical contribution is a governed multimodal epilepsy operating model — an articulation of clinical AI value as the product of detection quality, explanation legibility, and governance strength, and a demonstration that a stable governance core with evolvable model and retrieval complements is a viable structuring principle for clinical AI platforms. The methodological contribution is an end-to-end pipeline that combines real EEG, physiologically motivated and explainable features, and reproducible evaluation including external validation, showing how legible features can simultaneously drive performance and support clinical audit. The practical contribution is a deployable artefact comprising more than fifteen interactive analytical views together with a compiled dissertation, which makes the operating model inspectable by clinical, technical, and governance stakeholders alike. The artefact contribution is the concrete deliverable set — source code, documentation, and the compiled PDF — that instantiates the preceding three and provides a reproducible basis for extension.

Table 8.1. Contributions mapped to research questions.

Research question	Contribution class	Contribution	Principal evidence
RQ1: Can real-EEG seizure detection reach clinically credible, generalising performance?	Methodological	Reproducible real-EEG pipeline with explainable features and external validation	ROC-AUC 0.92 (CV), 0.97 (ictal-vs-interictal), 0.979 (external)
RQ2: Can such detection be made organisationally trustworthy and auditable?	Theoretical	Governed multimodal operating model; value as detection × explanation × governance	Governance-and-audit pipeline; per-inference attribution
RQ3: Can the capability be embedded in an adoptable enterprise structure?	Practical	Seven-pipeline platform with 15+ interactive views and compiled dissertation	Deployed viewer; documented pipeline contracts
RQ4: Can the whole be delivered reproducibly for extension?	Artefact	Code, documentation, and compiled PDF deliverable set	Repository and build outputs

■ *Diagram (Mermaid — interactive in the viewer): graph LR*

Figure 8.1. Contribution map. The four contribution classes reinforce one another and each answers a distinct research question, so that the artefact grounds the practical contribution, which realises the methodological one, which in turn evidences the theoretical claim.

8.3 Limitations

Intellectual honesty requires a frank account of the boundaries of what has been achieved, and these boundaries are substantial. First, only the EEG modality is real; the accompanying clinical cohort data — demographics, medication histories, and outcome labels used to exercise the multimodal and reasoning layers — are synthetic. The multimodal claims are therefore demonstrated as an architecture and a data contract rather than validated on genuine linked clinical records. Second, the deep-learning components are lightweight multilayer-perceptron stand-ins for the EEGNet, convolutional, and transformer architectures that the design anticipates; the reported performance is achieved with engineered features and modest classifiers, and the heavier models remain specified rather than trained. Third, the retrieval layer uses term-frequency–inverse-document-frequency (TF-IDF) representations rather than neural embeddings, so semantic retrieval is lexical and comparatively shallow. Fourth, the Model Context Protocol (MCP) and multi-agent orchestration layer is specified but not yet executable, meaning the autonomous coordination described in the design has not been run. Fifth, and most consequentially for clinical claims, the real EEG derives from a single subject; multi-subject, institutional review board (IRB)-approved clinical data are required before any statement about population-level generalisation can be sustained. Sixth, no prospective clinical trial has been conducted, so all performance figures are retrospective and offline. Table 8.2 pairs each limitation with a concrete mitigation and the future work that discharges it.

Table 8.2. Limitations mapped to mitigations and future work.

Limitation	Current mitigation	Future work that resolves it
Clinical cohort data are synthetic (only EEG is real)	Realistic data contracts and schema-faithful synthesis	Acquire IRB-approved, linked multimodal clinical records
Deep models are lightweight MLP stand-ins	Explainable engineered features already reach strong AUC	Train EEGNet/CNN/transformer models on spectrograms and scalograms
RAG uses TF-IDF, not neural embeddings	Provenance-linked retrieval over a curated knowledge graph	Introduce neural embeddings with a vector database
MCP / multi-agent layer specified, not executable	Interfaces and contracts fully defined	Implement executable MCP servers and agent orchestration
Single-subject real EEG	External-validation split within available data	Multi-subject, IRB-approved acquisition across sites
No prospective clinical trial	Retrospective external validation	Stepped-wedge and prospective validation studies

8.4 Future Work

The forward agenda follows directly from the limitations. The foundational step is the acquisition of IRB-approved, multi-subject, and genuinely multimodal clinical data, without which the population-level and multimodal claims cannot be tested. On such data, the anticipated deep architectures — EEGNet and convolutional models over spectrograms, and transformer models over scalograms — can be trained and compared against the current explainable baseline, ideally preserving attribution so that legibility survives the move to higher-capacity models. The retrieval layer can be advanced from TF-IDF to neural embeddings backed by a vector database, deepening semantic recall while retaining the knowledge-graph provenance that underwrites governance. The specified MCP and multi-agent layer can be implemented as executable services, and an online feature store can be introduced to support low-latency, streaming inference on incoming EEG. Culminating the agenda, the platform can be subjected to prospective and stepped-wedge clinical evaluation, the design of which is sketched in Figure 8.2.

■ *Diagram (Mermaid — interactive in the viewer): sequenceDiagram*

Figure 8.2. Envisioned clinical adoption pathway. Adoption proceeds through a governed sequence — pilot, shadow deployment, stepped-wedge rollout, and prospective validation — with the institutional review board and governance function gating each transition, so that clinical authority is extended to the platform only as evidence accumulates.

The sequence in Figure 8.2 is deliberately conservative. Shadow deployment allows the platform to run on live signal while all clinical action remains with the neurologist, generating concordance evidence at no patient risk. The stepped-wedge design then introduces the platform to successive units over time, yielding a rigorous estimate of its effect on review throughput and detection yield while every unit eventually receives the intervention. Only after prospective validation and governance sign-off does the platform assume any governed clinical role. This pathway is the operational expression of the governance-by-design philosophy argued throughout the dissertation.

8.5 Reflective Closing

This dissertation began from the conviction that the barrier to AI in remote epilepsy care is organisational as much as it is technical. The work bears that conviction out. A seizure detector that generalises to external data with a ROC-AUC near 0.979 is a genuine achievement, but its practical worth was shown to depend on the legibility of its features, the transparency of its explanations, and the strength of the governance surrounding it. By designing the capability into a governed, seven-pipeline operating model and delivering it as an inspectable artefact, this research offers a template for how health systems might adopt clinical AI responsibly rather than merely capably. The limitations are real and openly stated, and the future work is correspondingly concrete. What endures from the enquiry is a way of thinking about clinical AI value — as the product of performance, explanation, and governance — and a working demonstration that the three can be engineered together. It is offered in the hope that the next epilepsy platform is judged not only by how well it detects seizures, but by how well an organisation can trust, audit, and stand behind what it does.

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